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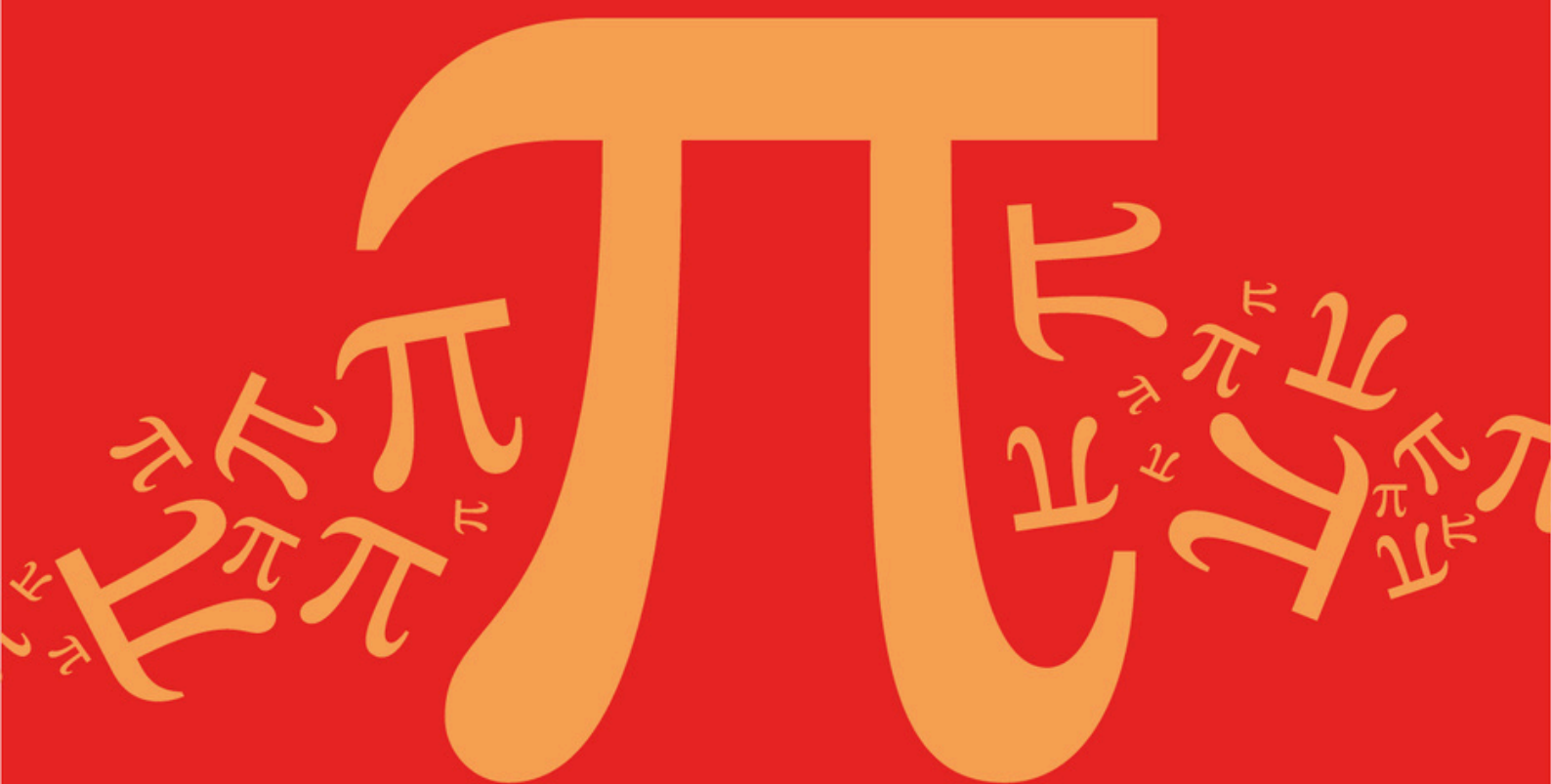
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AQA A Level Further Mathematics (7367)



Topic

DC: Network Flows

Sub topic

DC2: Understand Cuts

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **DC: Network Flows**
Sub topic **DC2: Understand Cuts**

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
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Name : _____

Date : _____

Time : _____

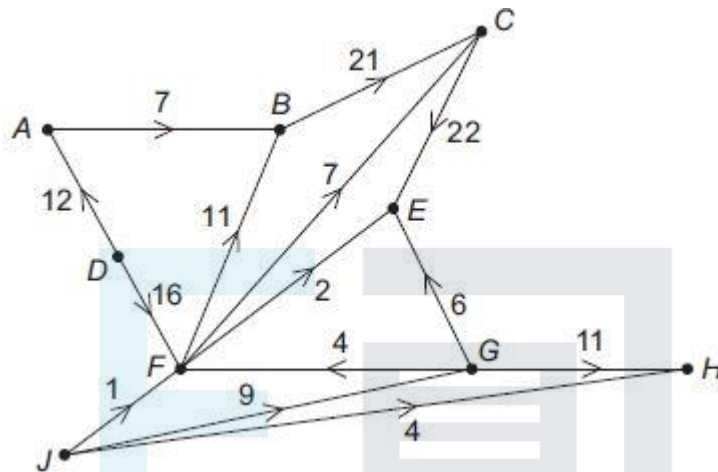
Total Marks Available : _____

Total Marks Received : _____

Q1.

The diagram shows a network of pipes.

Each pipe is labelled with its upper capacity in $\text{cm}^3 \text{ s}^{-1}$



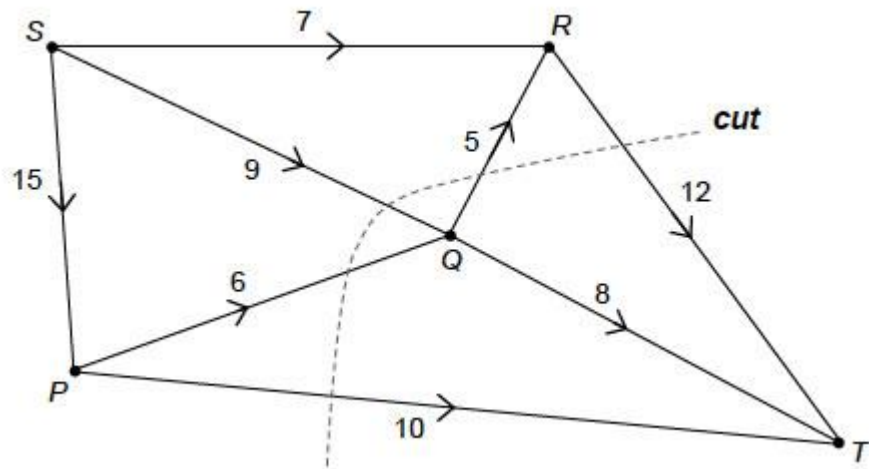
- (a) (i) Find the value of the cut given by $\{A, B, C, D, F, J\} \{E, G, H\}$. (1)
- (ii) State what can be deduced about the maximum flow through the network. (1)
- (b) (i) List the nodes which are sources of the network. (1)
- (ii) Add a supersource S to the network. (1)
- (c) (i) List the nodes which are sinks of the network. (1)
- (ii) Add a supersink T to the network. (1)
- (Total 6 marks)**

Q2.

The network shows a system of pipes, where S is the source and T is the sink.

The capacity, in litres per second, of each pipe is shown on each arc.

The cut shown in the diagram can be represented as $\{S, P, R\}, \{Q, T\}$.



- (a) Complete the table below to give the value of each of the 8 possible cuts.

Cut		Value
$\{S\}$	$\{P, Q, R, T\}$	31
$\{S, P\}$	$\{Q, R, T\}$	32
$\{S, Q\}$	$\{P, R, T\}$	
$\{S, R\}$	$\{P, Q, T\}$	
$\{S, P, Q\}$	$\{R, T\}$	30
$\{S, P, R\}$	$\{Q, T\}$	37
$\{S, Q, R\}$	$\{P, T\}$	35
$\{S, P, Q, R\}$	$\{T\}$	30

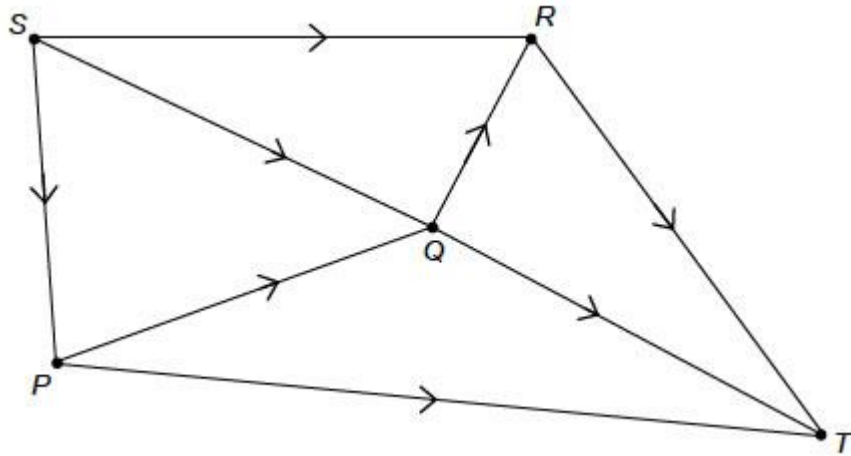
(1)

- (b) State the value of the maximum flow through the network.

Give a reason for your answer.

(1)

- (c) Indicate on the diagram a possible flow along each arc, corresponding to the maximum flow through the network.



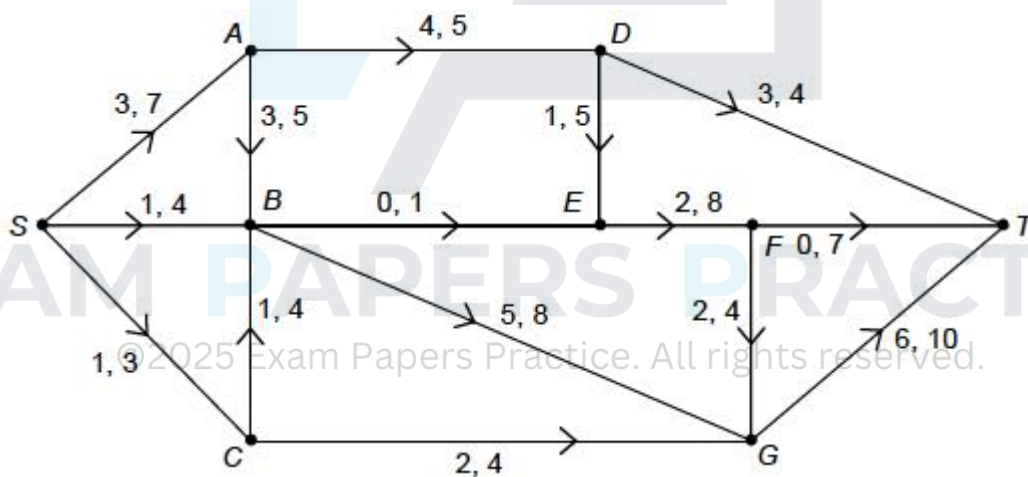
(2)

(Total 4 marks)

Q3.

The network shows a system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each arc.



(a) There is a feasible flow from S to T .

(i) Explain why arc AD must be at its lower capacity.

(1)

(ii) Explain why arc BE must be at its upper capacity.

(1)

(b) Explain why a flow of 11 litres per second through the network is impossible.

(1)

(c) The network in **Figure 1** shows a second system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each edge.

Figure 1

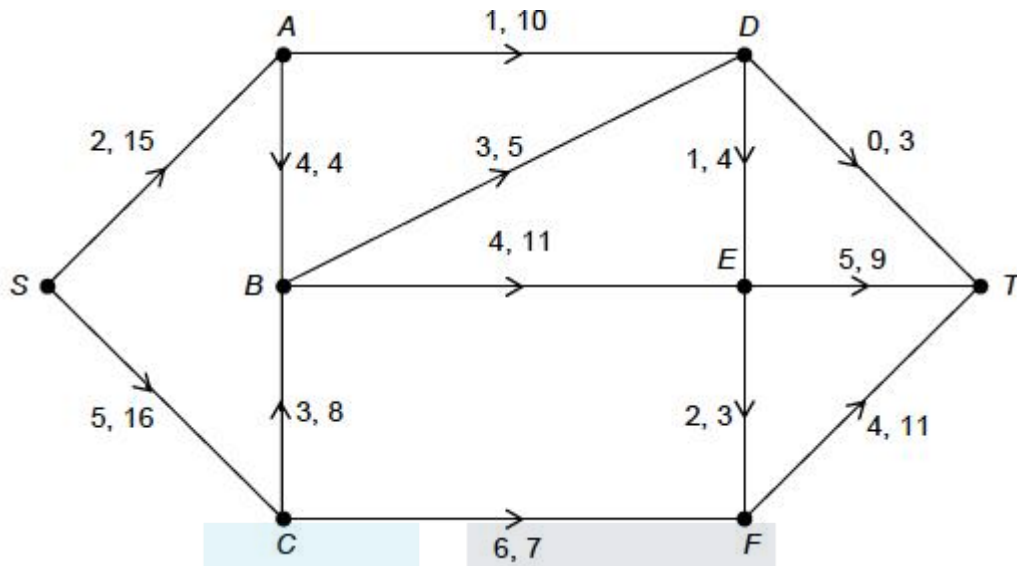
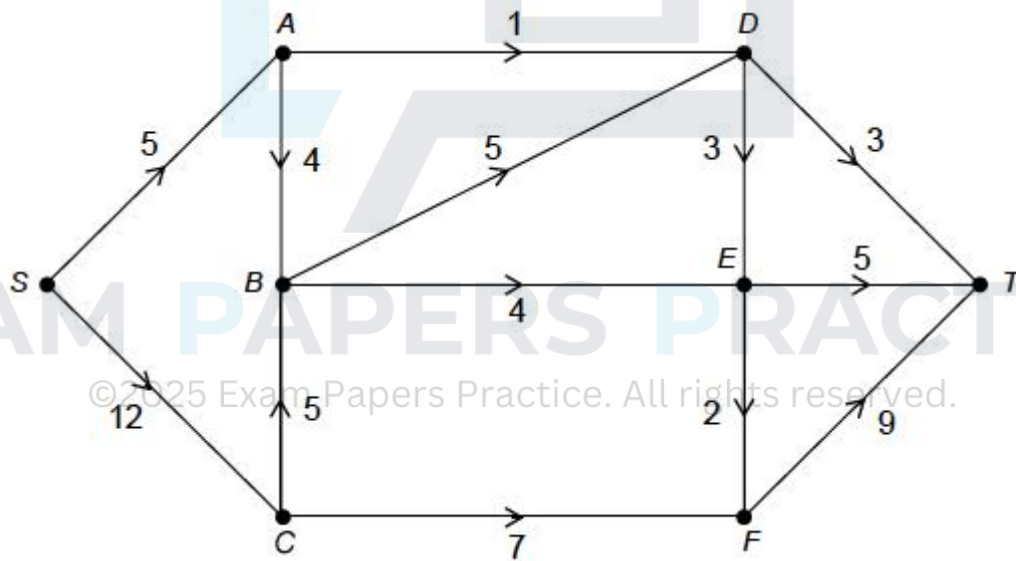


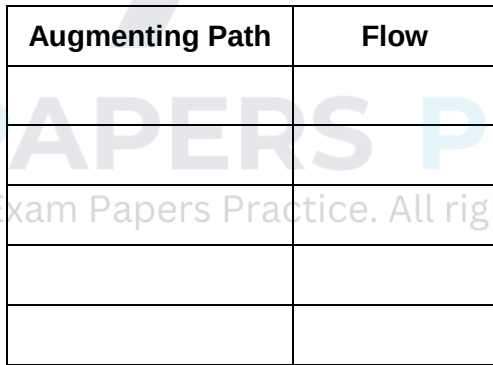
Figure 2 shows a feasible flow of 17 litres per second through the system of pipes.

Figure 2



- (i) Using **Figures 1** and **2**, indicate on **Figure 3** potential increases and decreases in the flow along each arc.

Figure 3



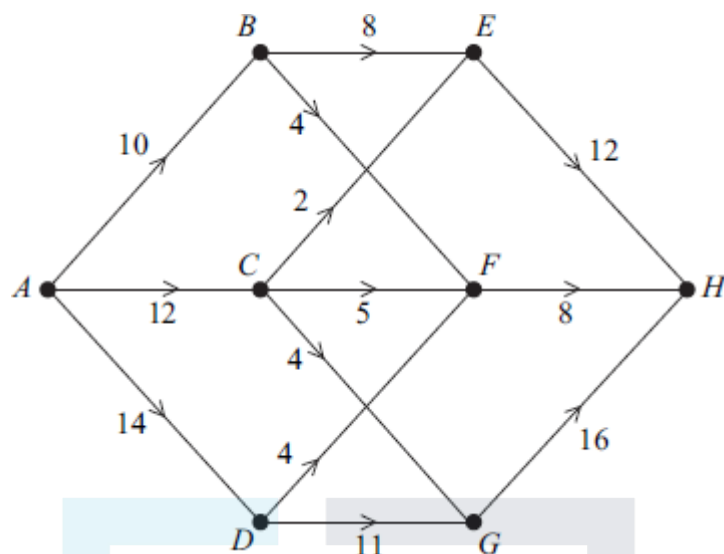
- You should indicate any flow augmenting paths clearly in the table below and modify the potential increases and decreases of the flow on **Figure 3**.
- (1)

Augmenting Path	Flow

- Interpret the impact of this restriction on the maximum flow through the system of pipes.

Q4.

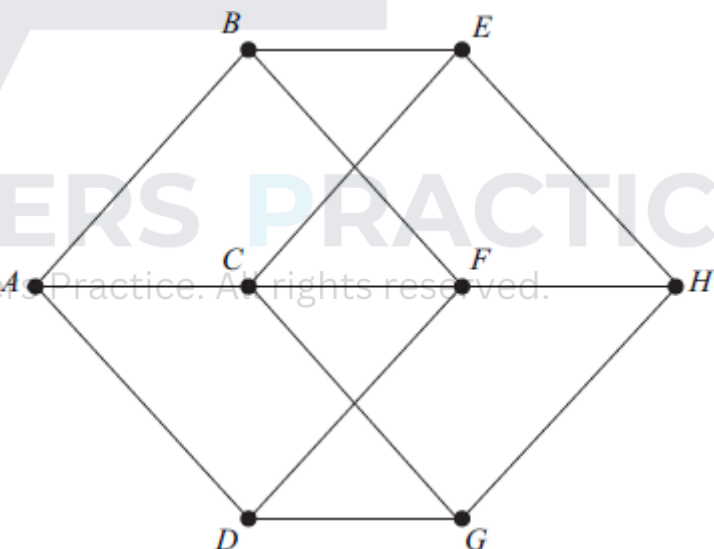
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- (a) Find the maximum flow along each of the routes $ABEH$, $ACFH$ and $ADGH$ and enter their values in the table on **Figure 1**.

Figure 1

Route	Flow
$ABEH$	
$ACFH$	
$ADGH$	



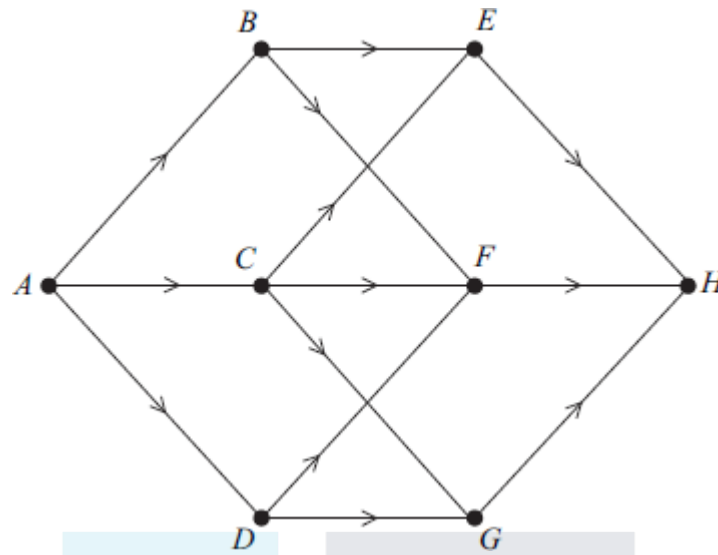
(1)

- (b) (i) Taking your answers to part (a) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow-augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (ii) State the value of the maximum flow and, on **Figure 2** below, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



(2)

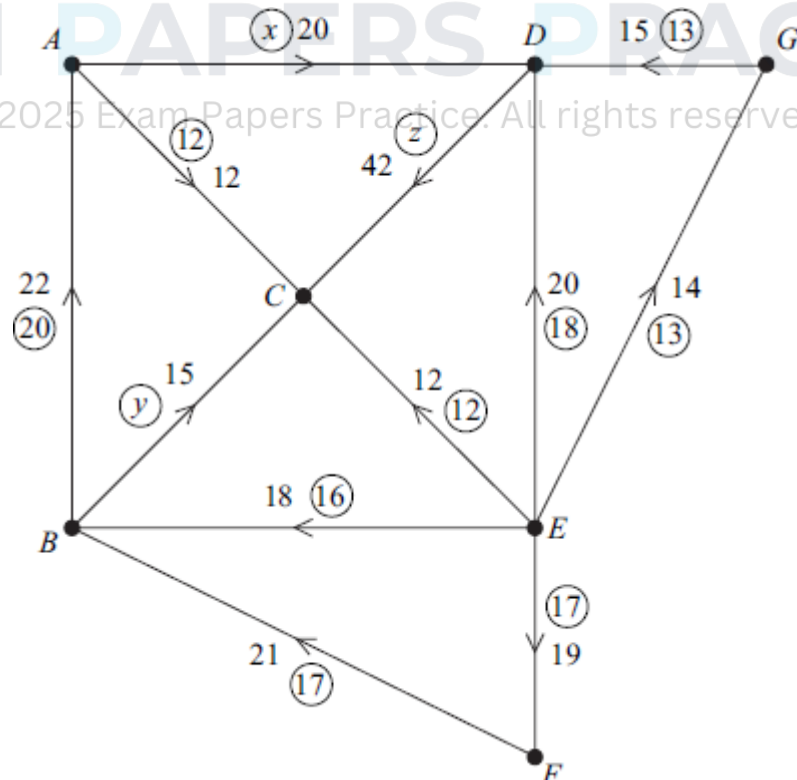
- (c) Confirm that you have a maximum flow by finding a cut of the same value. List the edges of your cut.

(1)

(Total 9 marks)

Q5.

The network below represents a system of pipes. The number **not** circled on each edge represents the capacity of each pipe in litres per second. The number or letter in each circle represents an initial flow in litres per second.



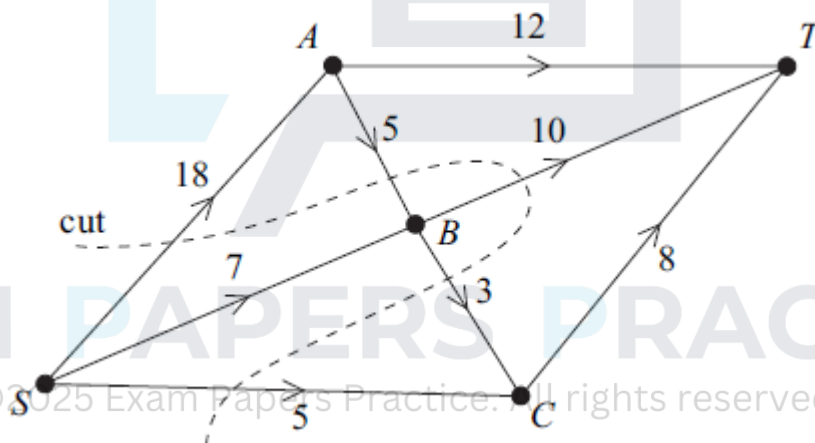
- (a) Write down the capacity of edge EF .

(1)

- (b) State the source vertex. (1)
- (c) State the sink vertex. (1)
- (d) Find the values of x , y and z . (3)
- (e) Find the value of the initial flow. (1)
- (f) Find the value of a cut through the edges EB , EC , ED , EF and EG . (1)
- (Total 8 marks)

Q6.

- (a) The network shows a flow from S to T along a system of pipes, with the capacity in litres per second indicated on each edge.



- (i) Show that the value of the cut shown on the diagram is 36 .

(1)

- (ii) The cut shown on the diagram can be represented as $\{S, B\}$, $\{A, C, T\}$.

Complete the table below to give the value of each of the 8 possible cuts.

Cut		Value
$\{S\}$	$\{A, B, C, T\}$	30
$\{S, A\}$	$\{B, C, T\}$	29
$\{S, B\}$	$\{A, C, T\}$	36
$\{S, C\}$	$\{A, B, T\}$	33

$\{S, A, B\}$	$\{C, T\}$	
$\{S, A, C\}$	$\{B, T\}$	
$\{S, B, C\}$	$\{A, T\}$	
$\{S, A, B, C\}$	$\{T\}$	30

(3)

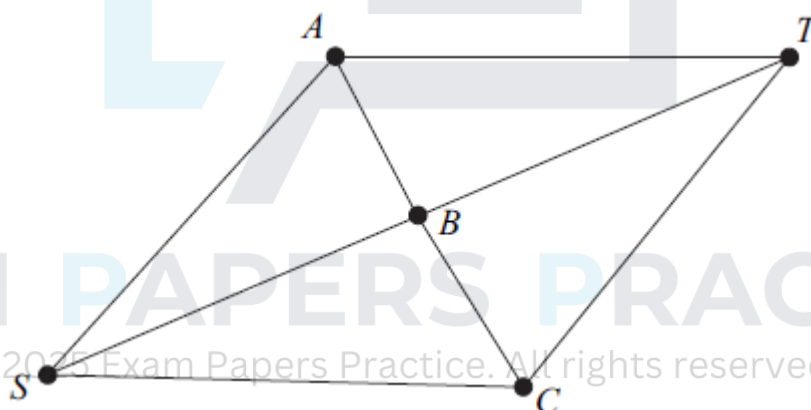
- (iii) State the value of the maximum flow through the network, giving a reason for your answer.

Maximum flow is _____

because _____

(2)

- (iv) Indicate on the diagram below a possible flow along each edge corresponding to this maximum flow.

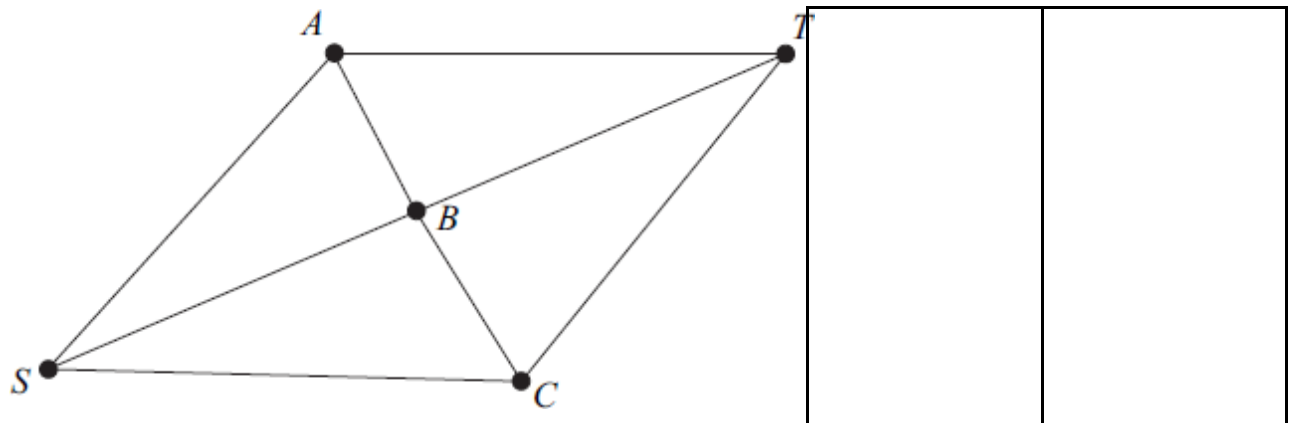


(1)

- (b) The capacities along SC and along AT are each increased by 4 litres per second.

- (i) Using your **values from part (a)(iv) as the initial flow**, indicate potential increases and decreases on the diagram below and use the labelling procedure to find the new maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the diagram.

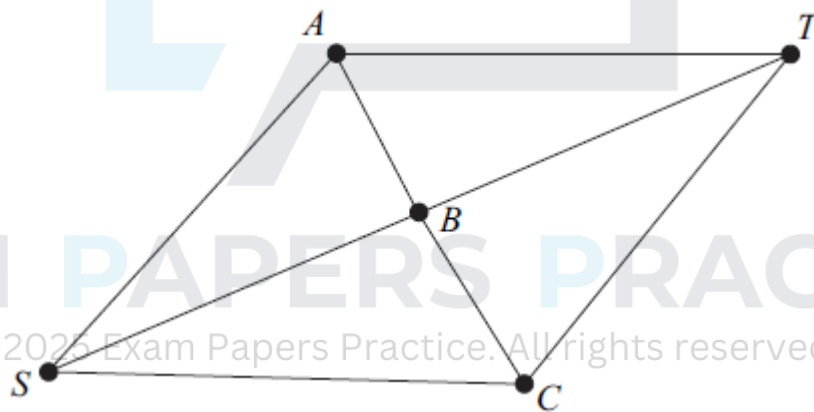
Path	Additional Flow



(6)

- (ii) Use your results from part (b)(i) to illustrate the flow along each edge that gives this new maximum flow, and state the value of the new maximum flow.

New maximum flow is _____

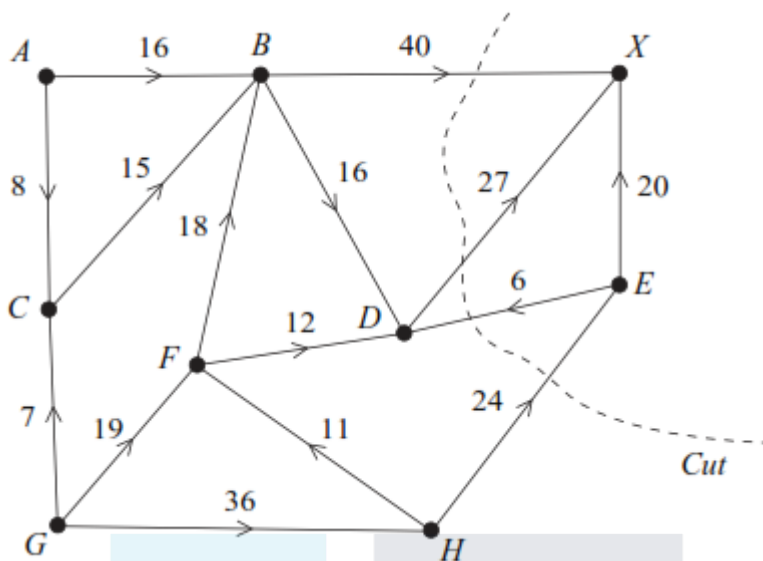


(3)

(Total 16 marks)

Q7.

The network shows the evacuation routes along corridors in a college, from two teaching areas to the exit, in case of a fire alarm sounding.



The two teaching areas are at A and G and the exit is at X.

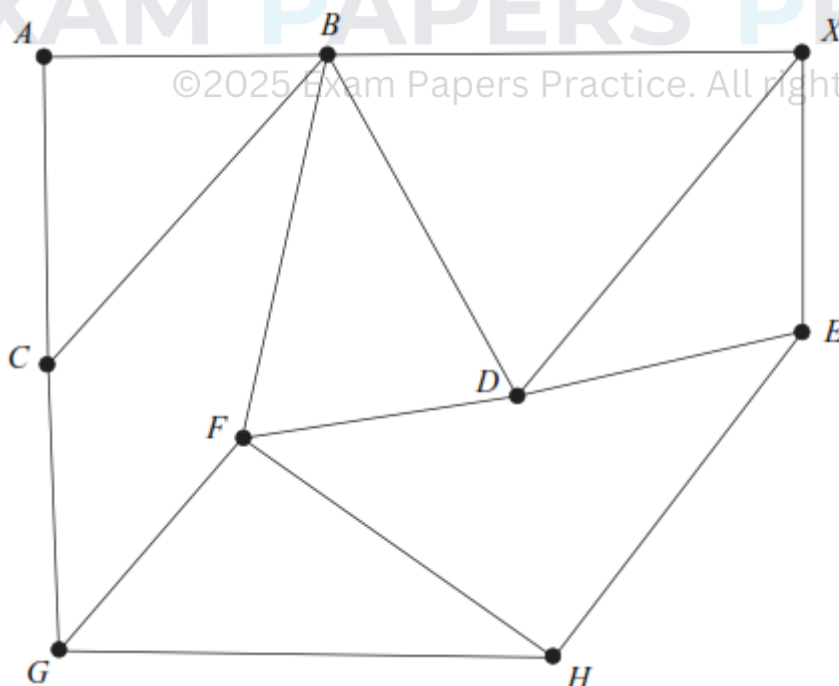
The number on each edge represents the maximum number of people that can travel along a particular corridor in one minute.

(a) Find the value of the cut shown on the diagram.

(1)

(b) Find the maximum flow along each of the routes $ABDX$, $GFBX$ and $GHEX$ and enter their values in the table on **Figure 1**.

Figure 1



Route	Flow
$ABDX$	
$GFBX$	
$GHEX$	

(3)

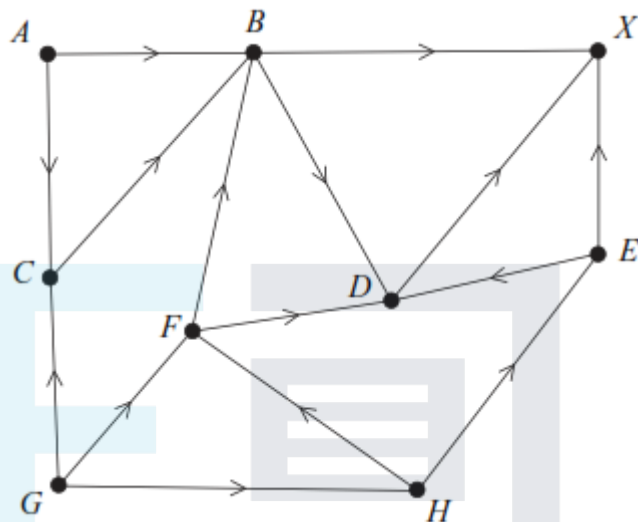
(c) (i) Taking your answers to part (b) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should

indicate any flow augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



Maximum flow is _____ people per minute.

(2)

- (d) During one particular fire drill, there is an obstruction allowing no more than 45 people per minute to pass through vertex B . State the maximum number of people that can move through the network per minute during this fire drill.

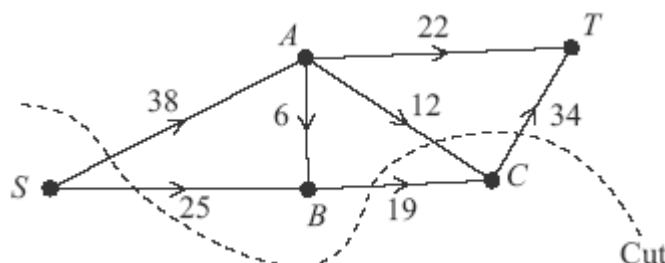
(2)

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(Total 13 marks)

Q8.

- (a) The network shows a flow from S to T along a system of pipes, with the capacity, in litres per minute, indicated on each edge.



- (i) Show that the value of the cut shown on the diagram is 97.

(1)

- (ii) The cut shown on the diagram can be represented as $\{S, C\}, \{A, B, T\}$.

Complete the table on **Figure 1**, giving the value of each of the 8 possible cuts.

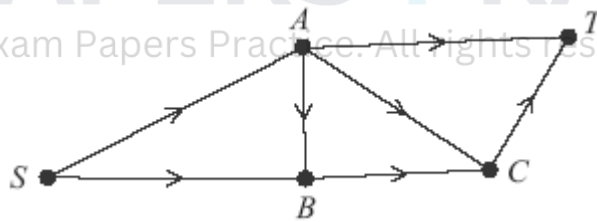
Figure 1

Cut		Value
$\{S\}$	$\{A, B, C, T\}$	63
$\{S, A\}$	$\{B, C, T\}$	
$\{S, B\}$	$\{A, C, T\}$	
$\{S, C\}$	$\{A, B, T\}$	97
$\{S, A, B\}$	$\{C, T\}$	53
$\{S, A, C\}$	$\{B, T\}$	87
$\{S, B, C\}$	$\{A, T\}$	
$\{S, A, B, C\}$	$\{T\}$	

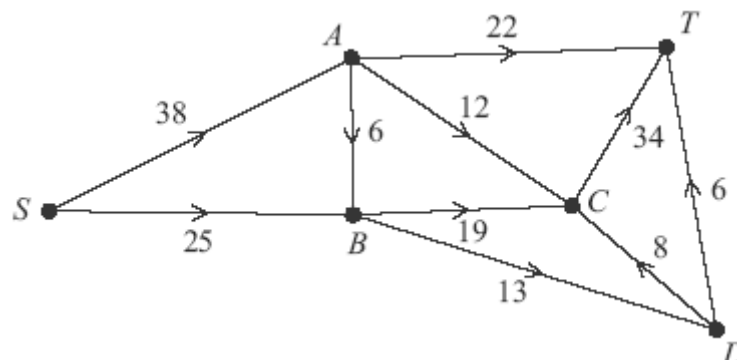
- (iii) State the value of the maximum flow through the network, giving a reason for your answer.

- (iv) Indicate on **Figure 2** a possible flow along each edge corresponding to this maximum flow.

Figure 2



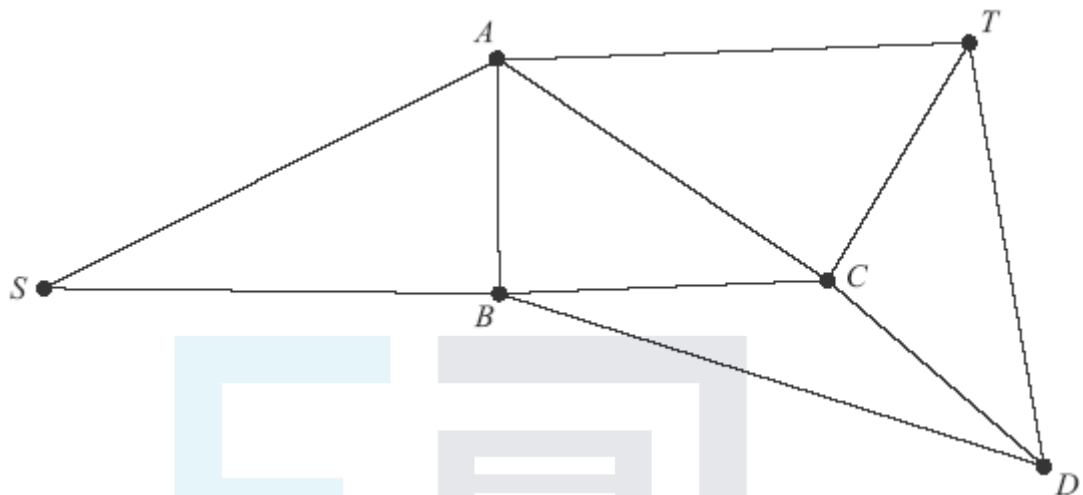
- (b) Extra pipes, BD , CD and DT , are added to form a new system, with the capacity, in litres per minute, indicated on each edge of the network below.



- (i) Taking your values from Figure 2 as the initial flow, use the labelling

procedure on Figure 3 to find the new maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 3



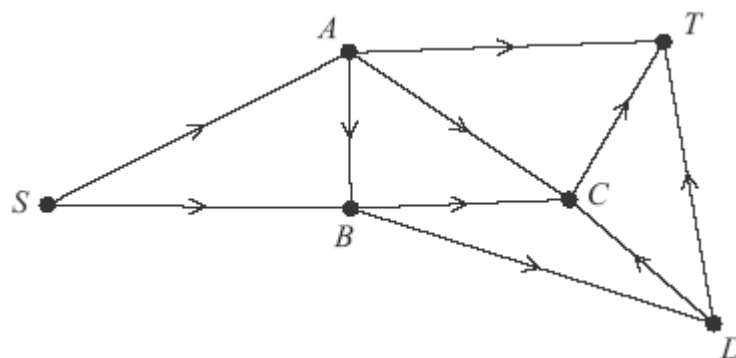
Path	Additional flow

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(4)

- (ii) State the value of the new maximum flow, and, on **Figure 4**, indicate a possible flow along each edge corresponding to this maximum flow.

Figure 4



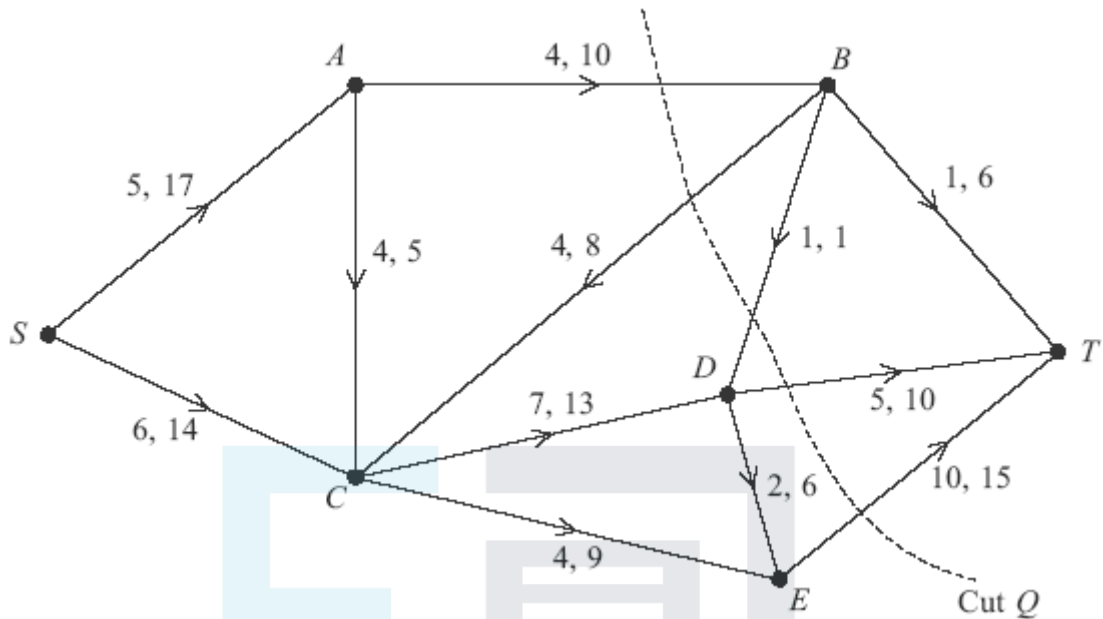
New maximum flow is _____

(2)

(Total 15 marks)

Q9.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per minute.

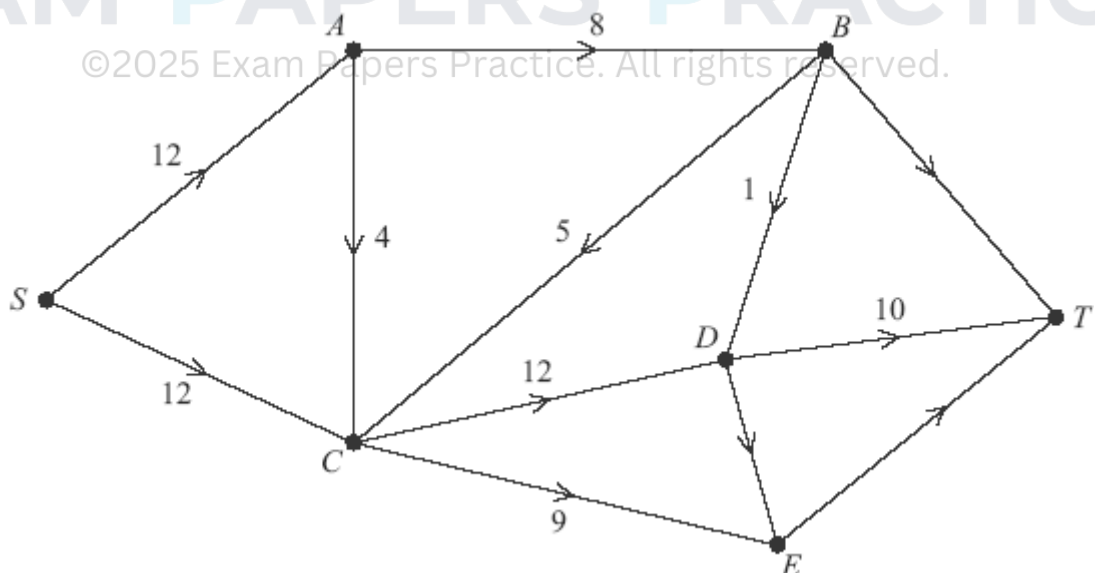


- (a) Find the value of the cut Q .

(2)

- (b) **Figure 1** below shows a partially completed diagram for a feasible flow of 24 litres per minute from S to T . Indicate, on **Figure 1**, the flows along the edges BT , DE and ET .

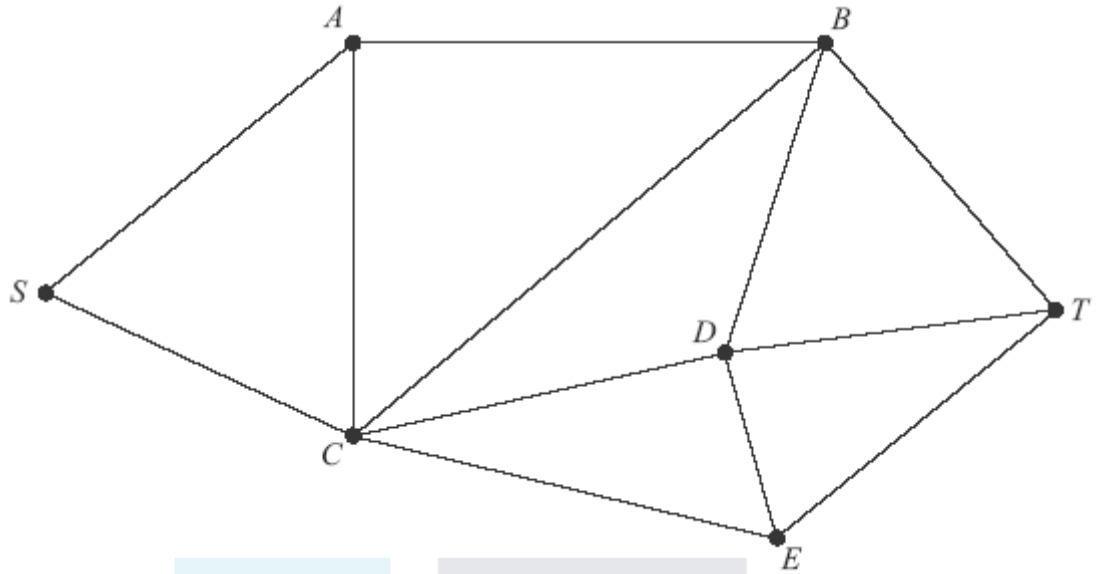
Figure 1



(2)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2** below.

Figure 2



Path	Flow

(2)

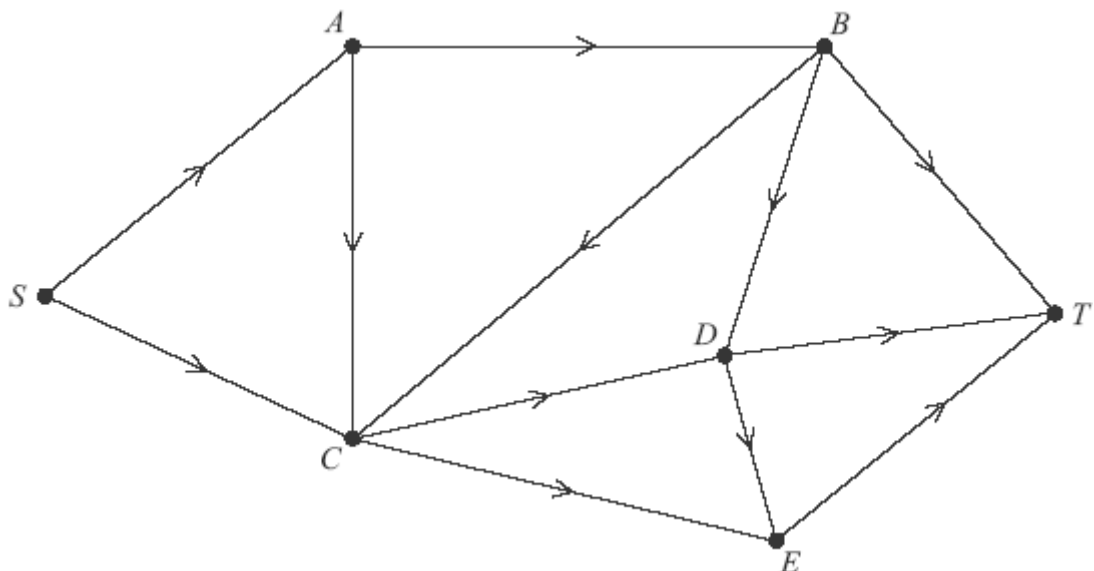
(ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T .

You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

(iii) Illustrate the maximum flow on **Figure 3** below.

Figure 3



(2)

- (d) Find a cut with value equal to that of the maximum flow.

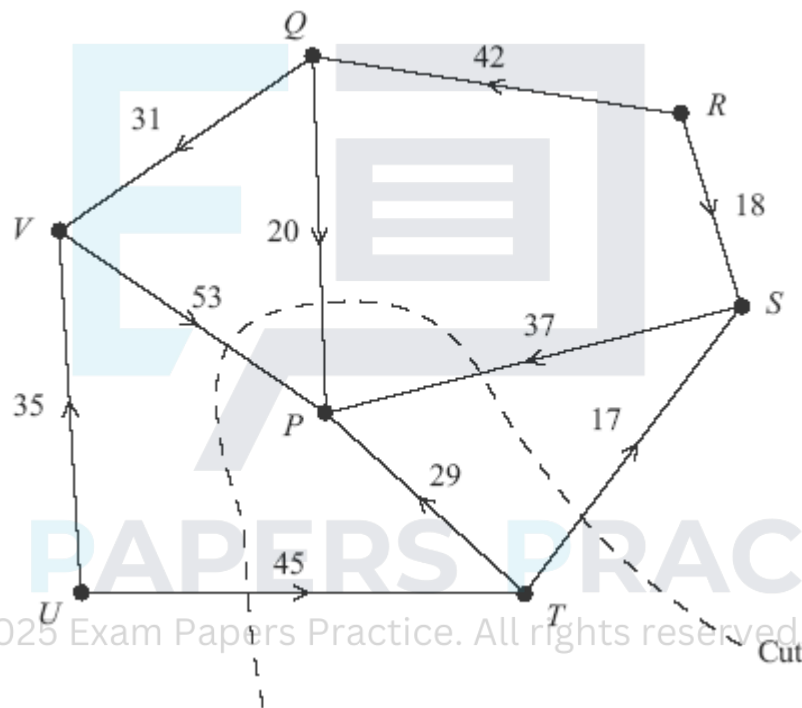
You may wish to show the cut on the network above.

(1)

(Total 14 marks)

Q10.

The network shows the routes along corridors from two arrival gates to the passport control area, P , in a small airport. The number on each edge represents the maximum number of passengers that can travel along a particular corridor in one minute.



- (a) State the vertices that represent the arrival gates.

(1)

- (b) Find the value of the cut shown on the network.

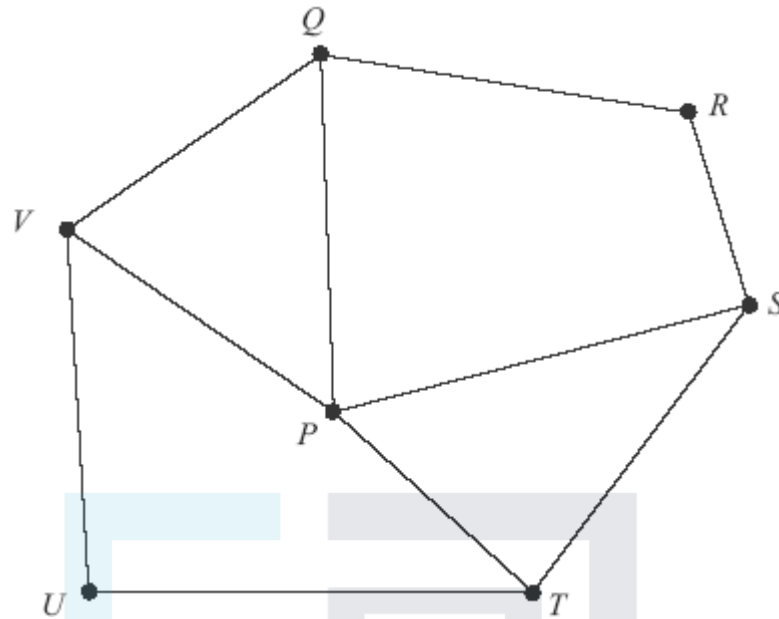
(1)

- (c) State the maximum flow along each of the routes $UTSP$ and $RQVP$.

(2)

- (d) (i) Taking your answers to part (c) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 1



Route	Value of Flow
<i>UTSP</i>	
<i>RQVP</i>	

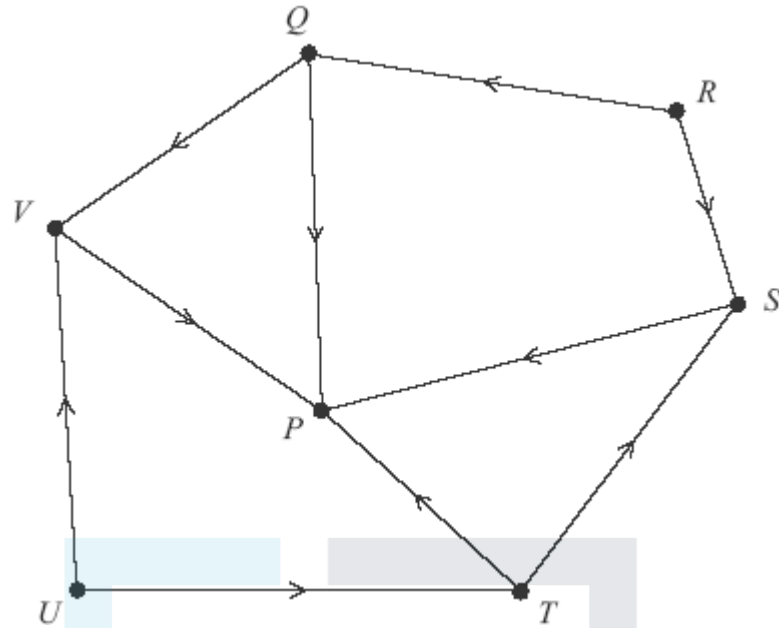
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(6)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2

Maximum flow is _____ passengers per minute



(2)

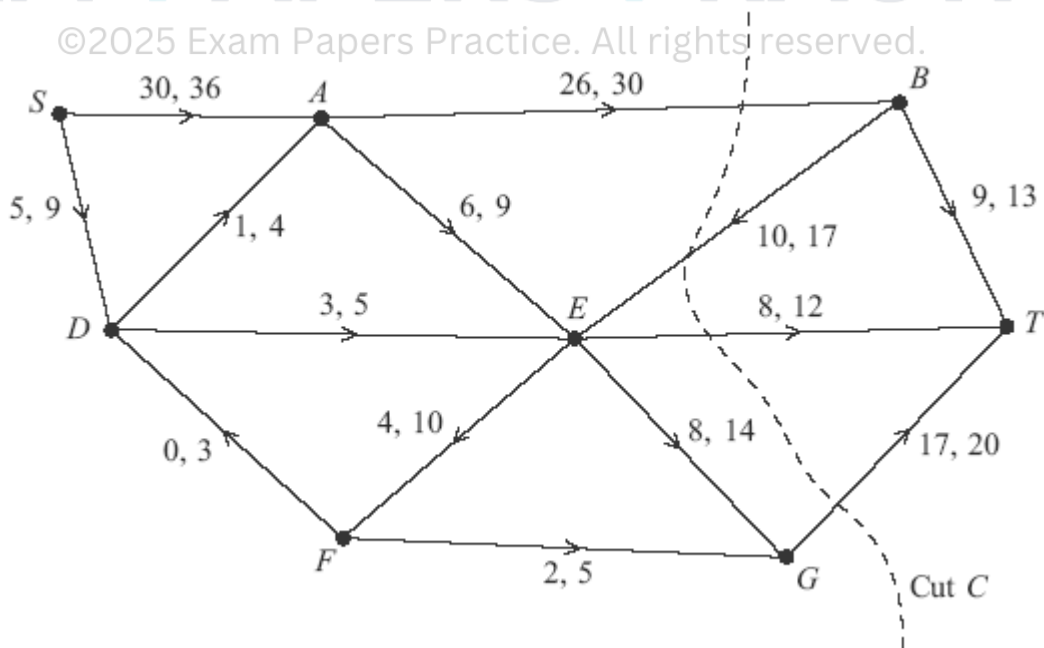
- (e) On a particular day, there is an obstruction allowing no more than 50 passengers per minute to pass through vertex V . State the maximum number of passengers that can move through the network per minute on this particular day.

(2)

(Total 14 marks)

Q11.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



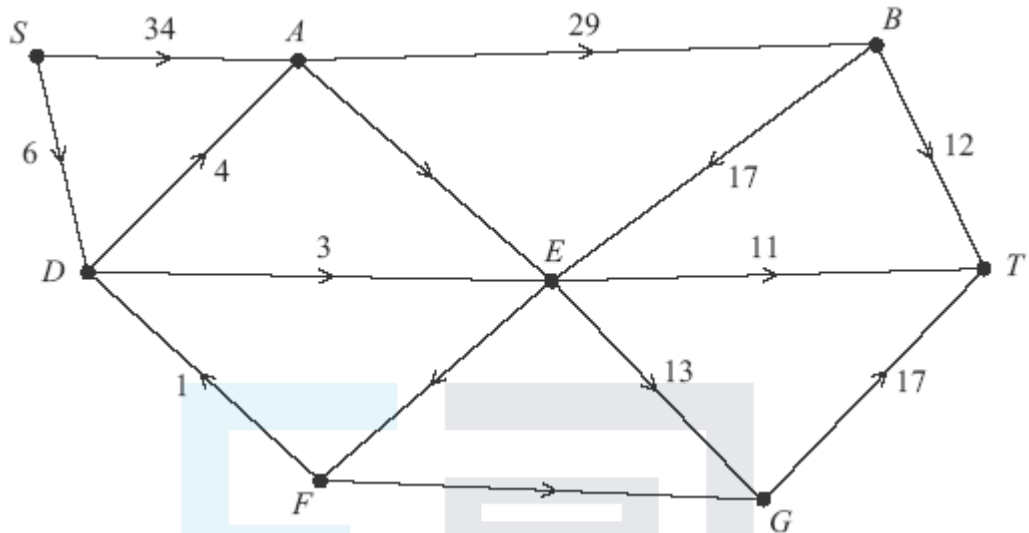
- (a) Find the value of the cut C .

(2)

- (b) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 40 litres

per second from S to T . Indicate, on **Figure 1**, the flows along the edges AE , EF and FG .

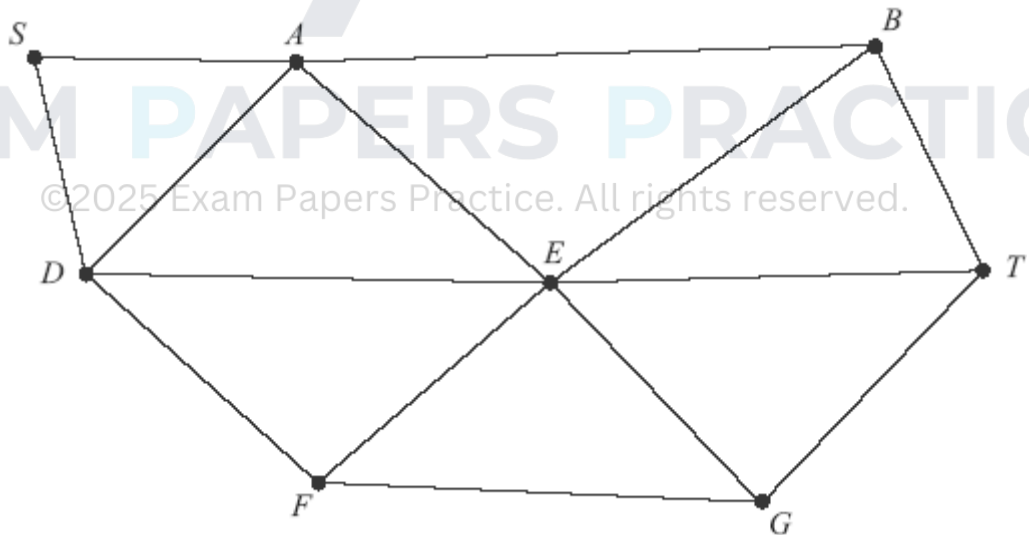
Figure 1



(3)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Extra Flow

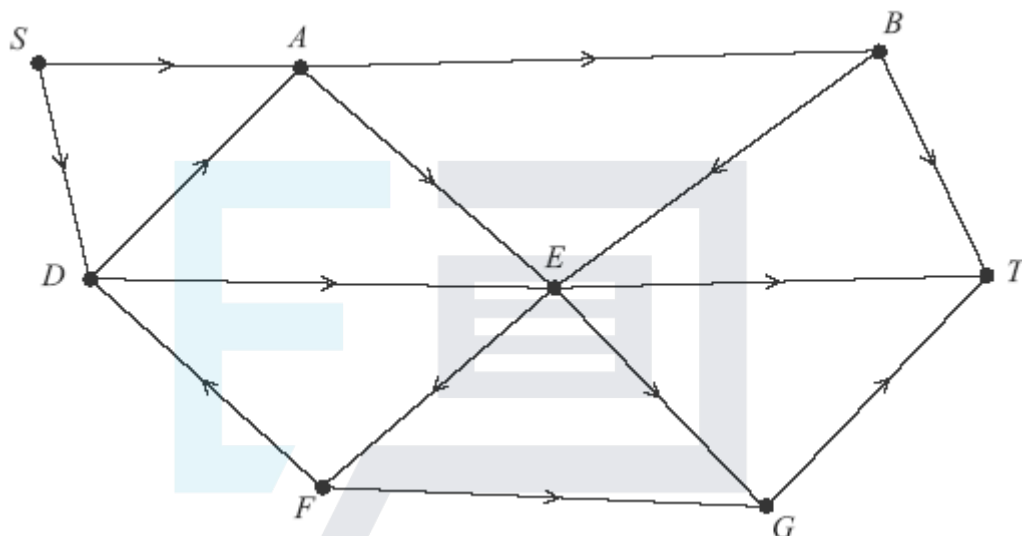
(3)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(4)

- (d) Illustrate the maximum flow on **Figure 3**.

Figure 3



(2)

- (e) Find a cut with value equal to that of the maximum flow.

(2)

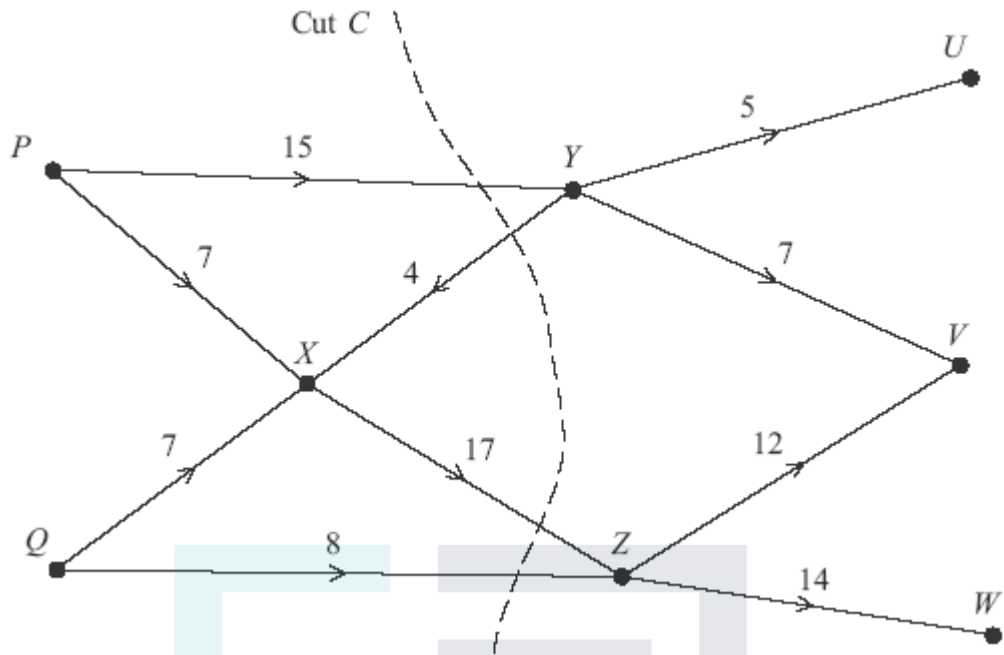
(Total 16 marks)

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Q12.

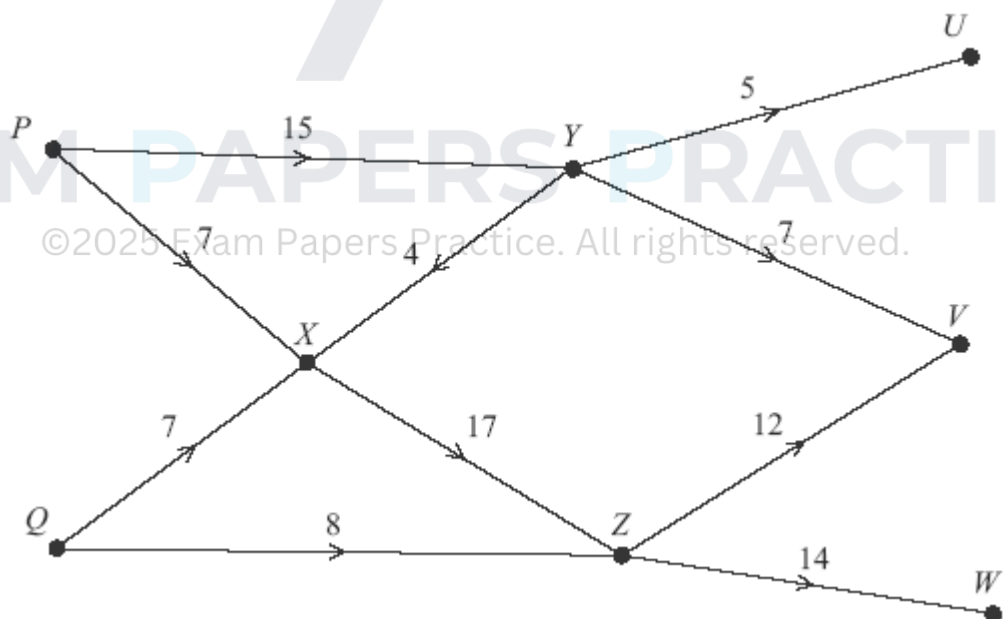
Water has to be transferred from two mountain lakes P and Q to three urban reservoirs U , V and W . There are pumping stations at X , Y and Z .

The possible routes with the capacities along each edge, in millions of litres per hour, are shown in the following diagram.



- (a) On **Figure 1**, add a super-source, S , and a super-sink, T , and appropriate edges so as to produce a directed network with a single source and a single sink. Indicate the capacity of each of the edges you have added.

Figure 1



- (b) (i) Find the value of the cut C . (2)
- (ii) State what can be deduced about the maximum flow from S to T . (1)
- (c) On **Figure 2**, write down the maximum flows along the routes $SQZWT$ and $SPYXZVT$. (1)

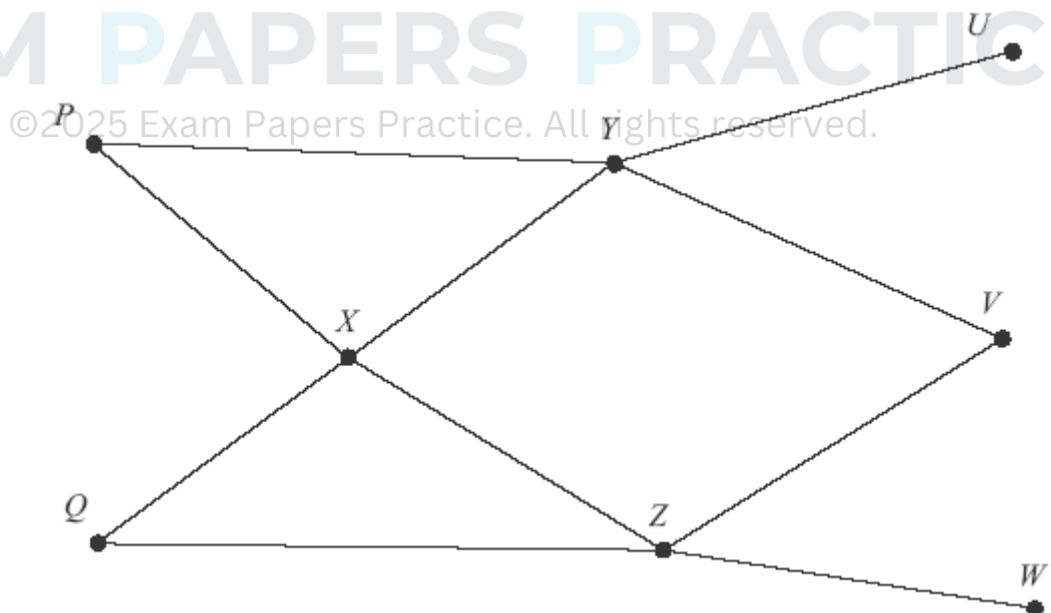
Figure 2

Route	Flow
<i>SQZWT</i>	
<i>SPYXZVT</i>	

(2)

- (d) (i) On **Figure 3**, add the vertices *S* and *T* and the edges connecting *S* and *T* to the network. Using the maximum flows along the routes *SQZWT* and *SPYXZVT* found in part (c) as the initial flow, indicate the potential increases and decreases of flow on each edge.

Figure 3



(2)

- (ii) Use flow augmentation to find the maximum flow from *S* to *T*. You should indicate any flow augmenting paths on **Figure 2** and modify the potential increases and decreases of the flow on **Figure 3**.

(4)

- (e) State the value of the flow from *Y* to *X* in millions of litres per hour when the

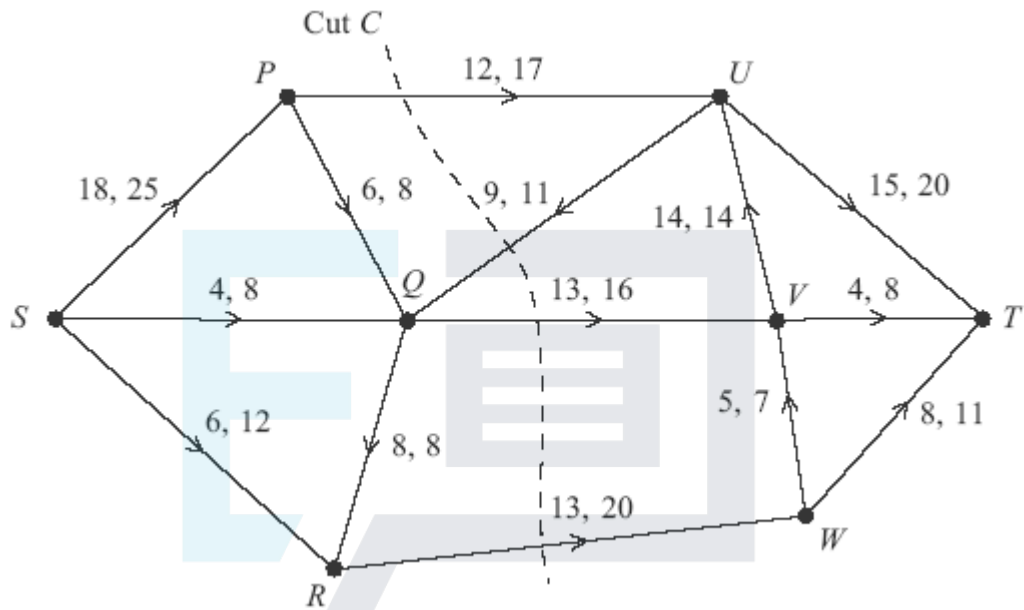
maximum flow is achieved.

(1)

(Total 13 marks)

Q13.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



- (a) (i) Find the value of the cut C .

(1)

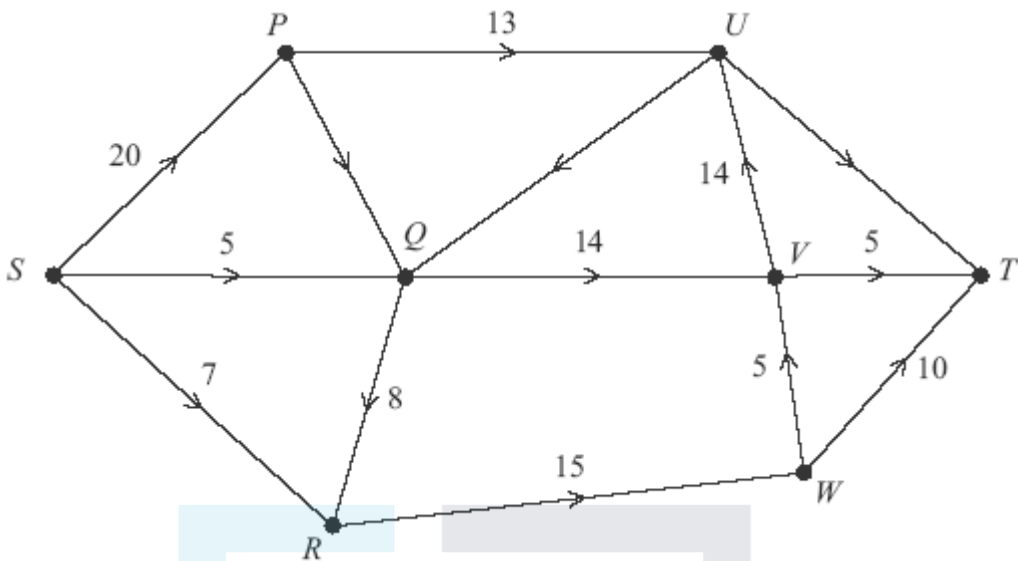
- (ii) Hence state what can be deduced about the maximum flow from S to T .

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(1)

- (b) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 32 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges PQ , UQ and UT .

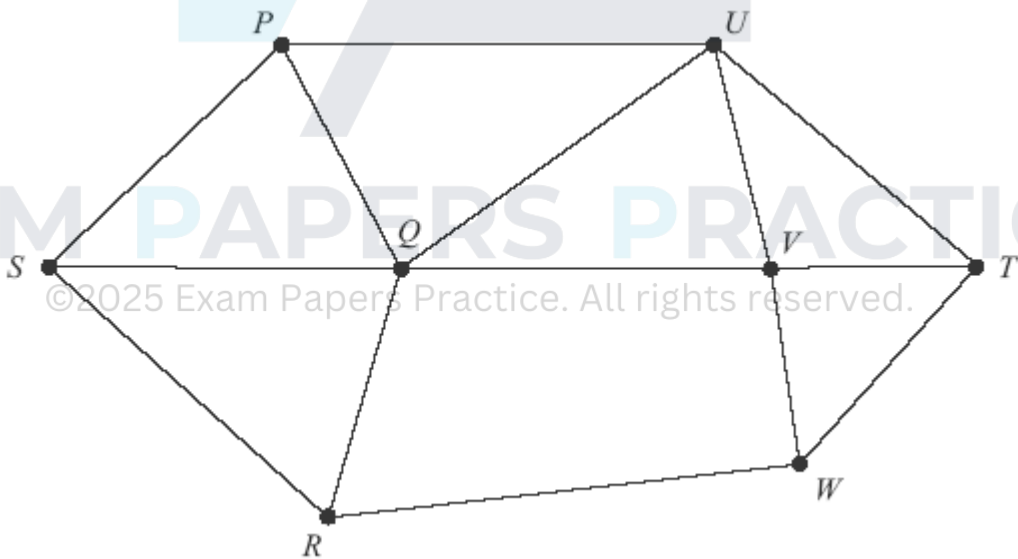
Figure 1



(3)

- (c) (i) Taking your feasible flow from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Additional Flow

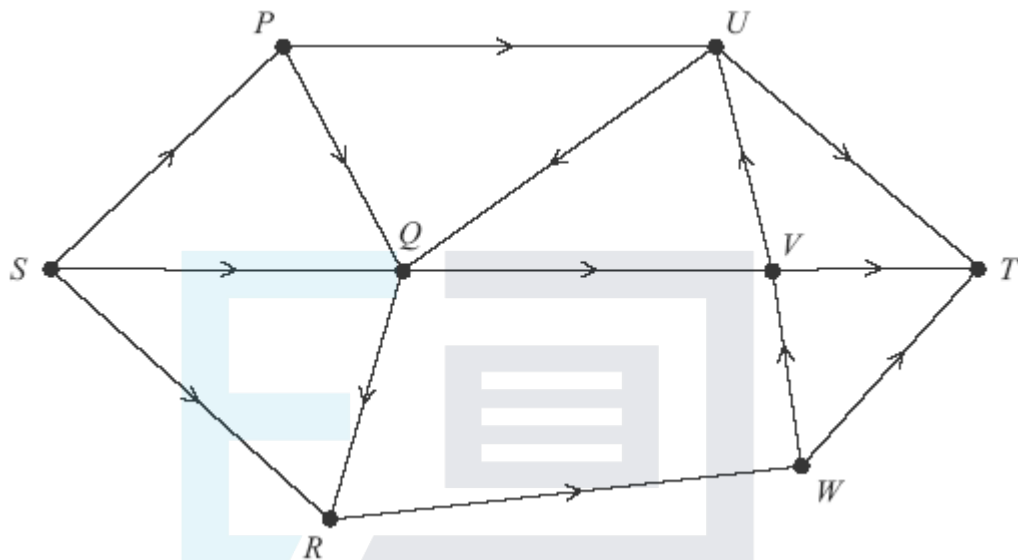
(2)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (iii) Illustrate the maximum flow on **Figure 3**.

Figure 3

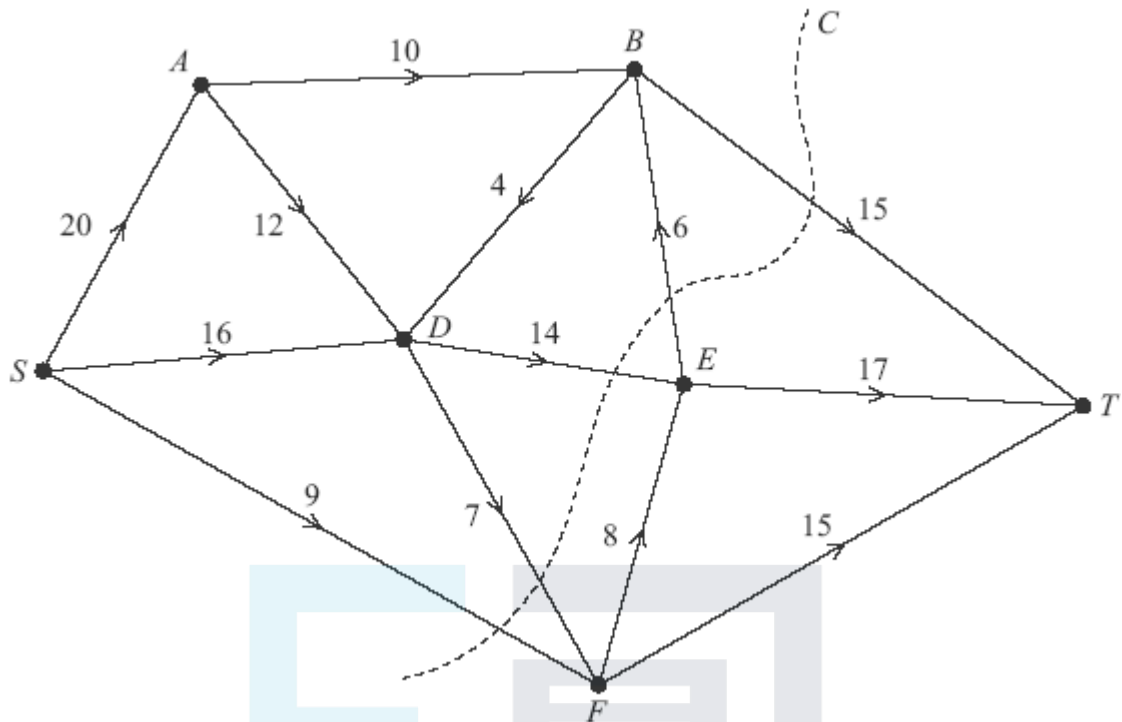


(1)

(Total 13 marks)

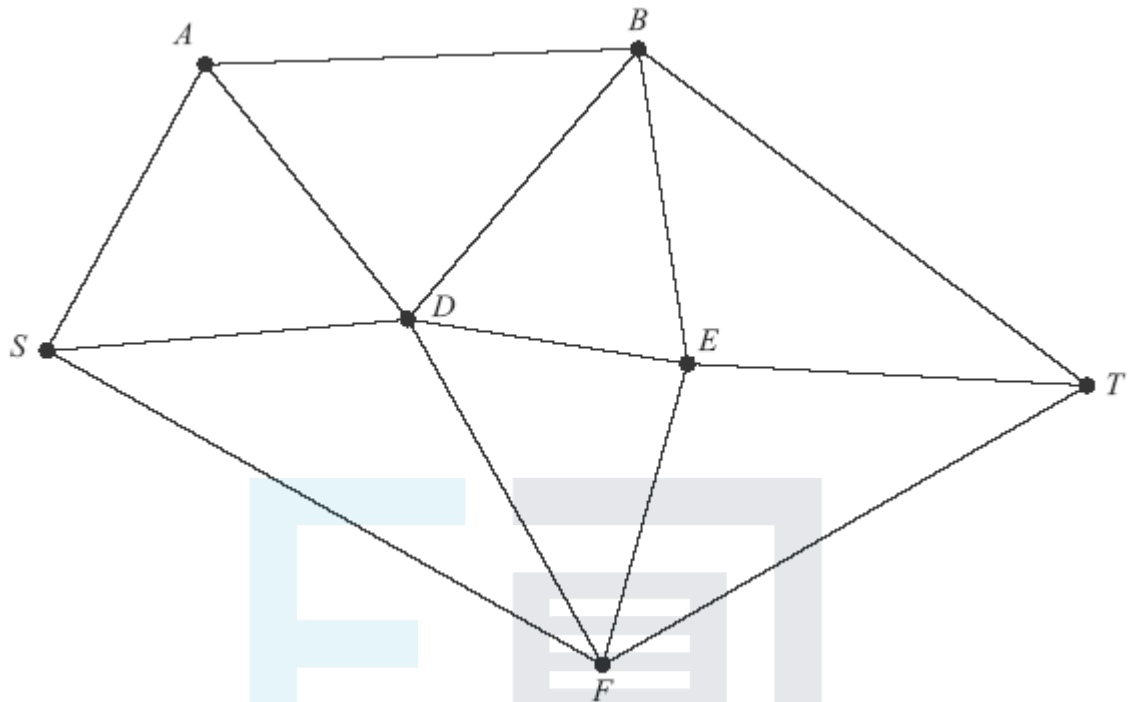
Q14.

The diagram shows a network of pipelines through which oil can travel. The oil field is at S , the refinery is at T and the other vertices are intermediate stations. The weights on the edges show the capacities in millions of barrels per hour that can flow through each pipeline.



- (a) (i) Find the value of the cut marked C on the diagram. (1)
- (ii) Hence make a deduction about the maximum flow of oil through the network. (2)
- (b) State the maximum possible flows along the routes $SABT$, $SDET$ and SFT . (2)
- (c) (i) Taking your answer to part (b) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow from S to T . Record your routes and flows in the table provided and show the augmented flows on the network diagram.

Figure 1



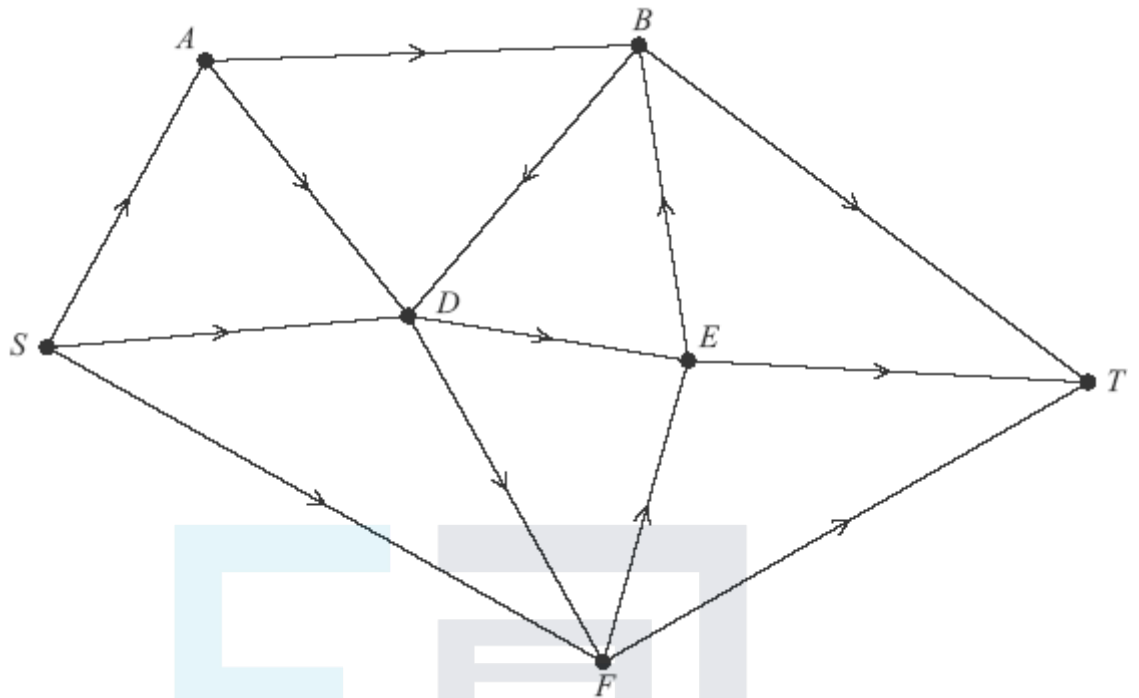
Route	Flow
<i>SABT</i> <i>SDET</i> <i>SFT</i>	

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(6)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



(2)

(iii) Prove that your flow in part (c)(ii) is a maximum.

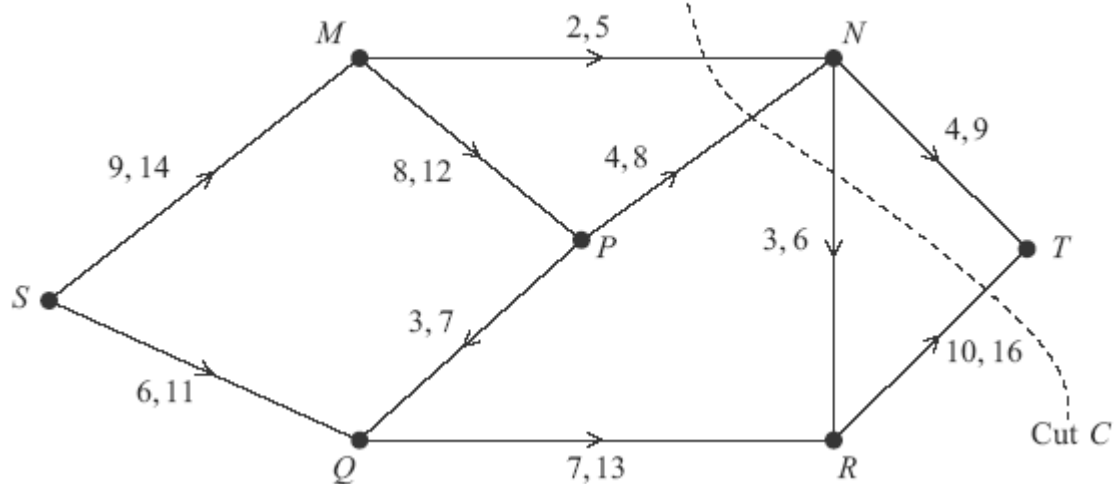
(2)

(Total 15 marks)

Q15.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.

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(a) (i) Find the value of the cut C .

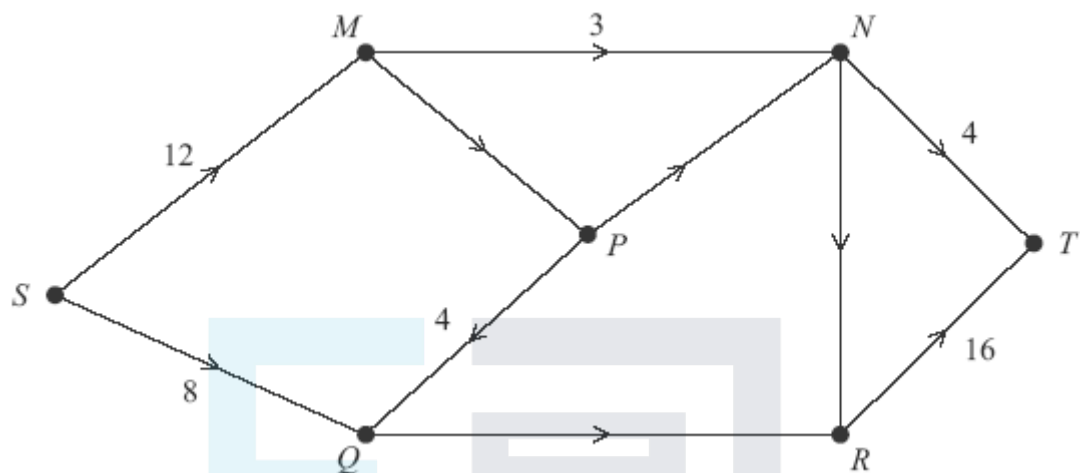
(1)

(ii) State what can be deduced about the maximum flow from S to T .

(1)

- (b) **Figure 1** shows a partially completed diagram for a feasible flow of 20 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MP , PN , QR and NR .

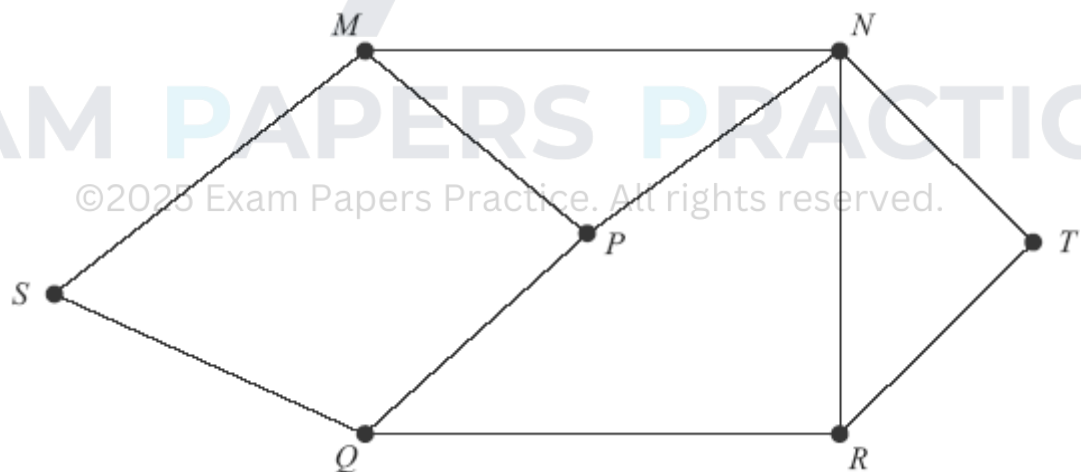
Figure 1



(4)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Flow

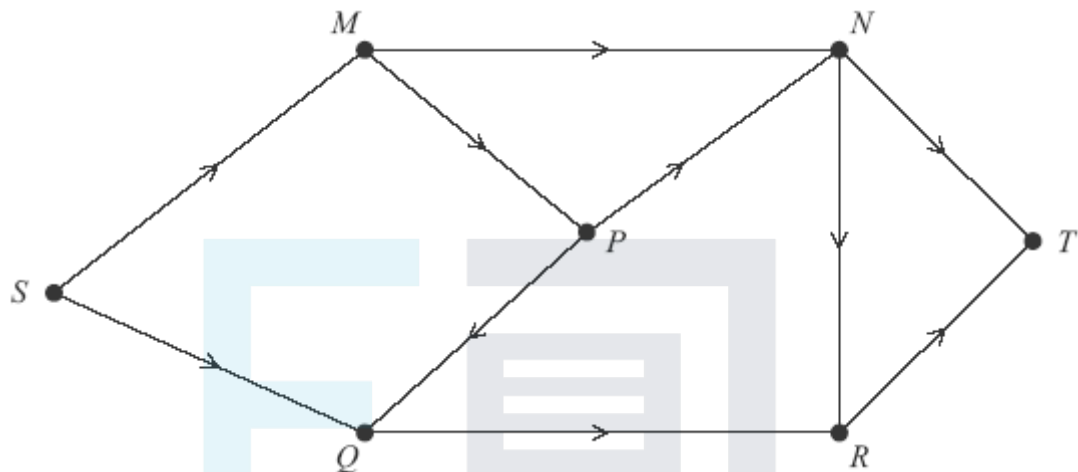
(2)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (iii) Illustrate the maximum flow on **Figure 3**.

Figure 3

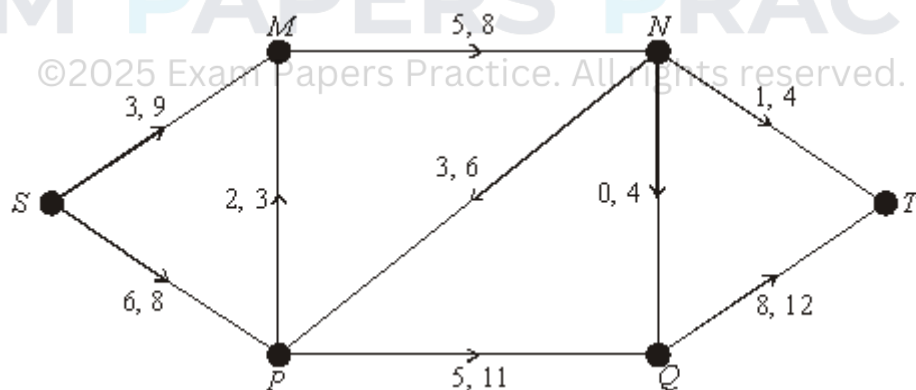


(2)

(Total 15 marks)

Q16.

The network shows a system of pipes, with the lower and upper capacities for each pipe in litres per second.



- (a) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 10 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MN , PQ , NP and NT .

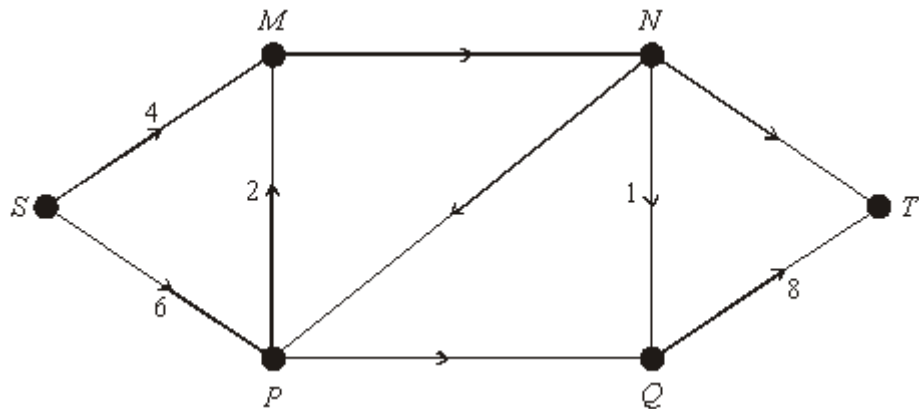
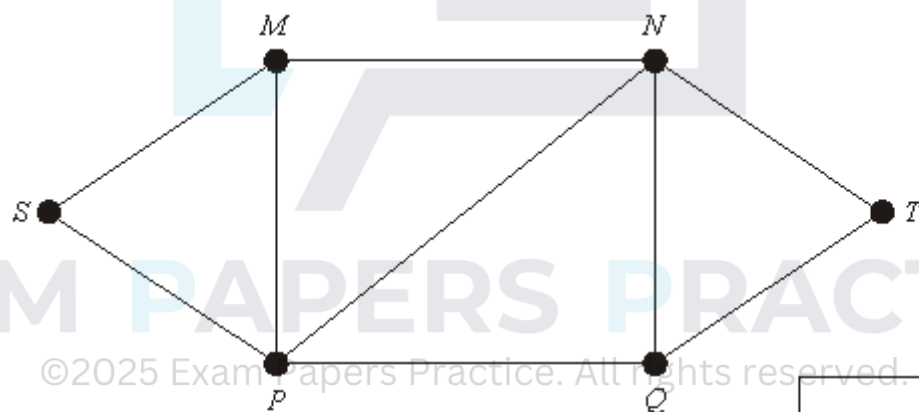


Figure 1

(4)

- (b) (i) Taking your answer from part (a) as an initial flow, use flow augmentation on **Figure 2** to find the maximum flow from S to T .



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Route	Extra flow

Figure 2

(6)

- (ii) State the value of the maximum flow and illustrate this flow on **Figure 3**.

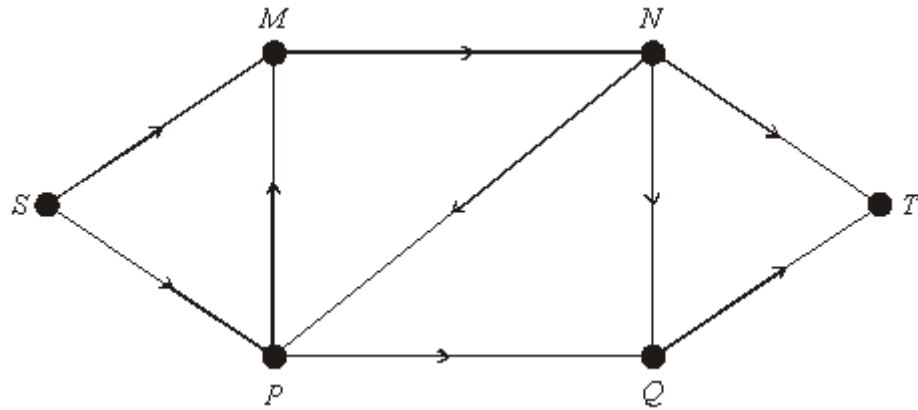


Figure 3

(2)

- (c) Find a cut with capacity equal to that of the maximum flow.

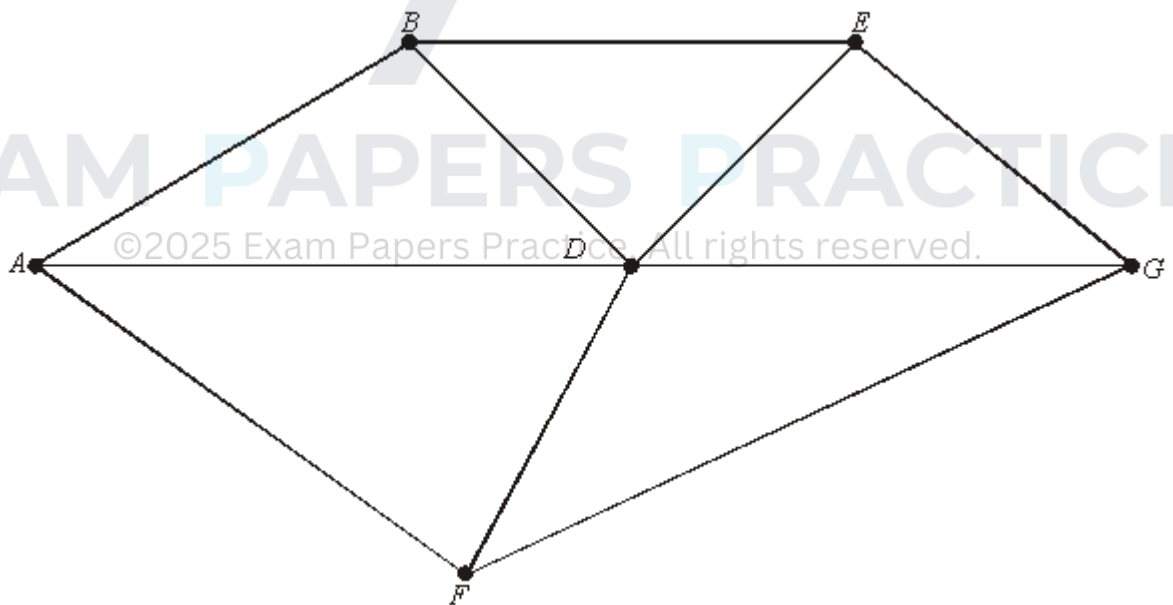
(2)

(Total 14 marks)

Q17.

[Figures 1 and 2, below, are provided for use in this question.]

Figure 1



Route	Flow
ABC	
GED	

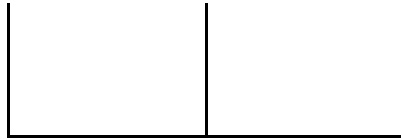
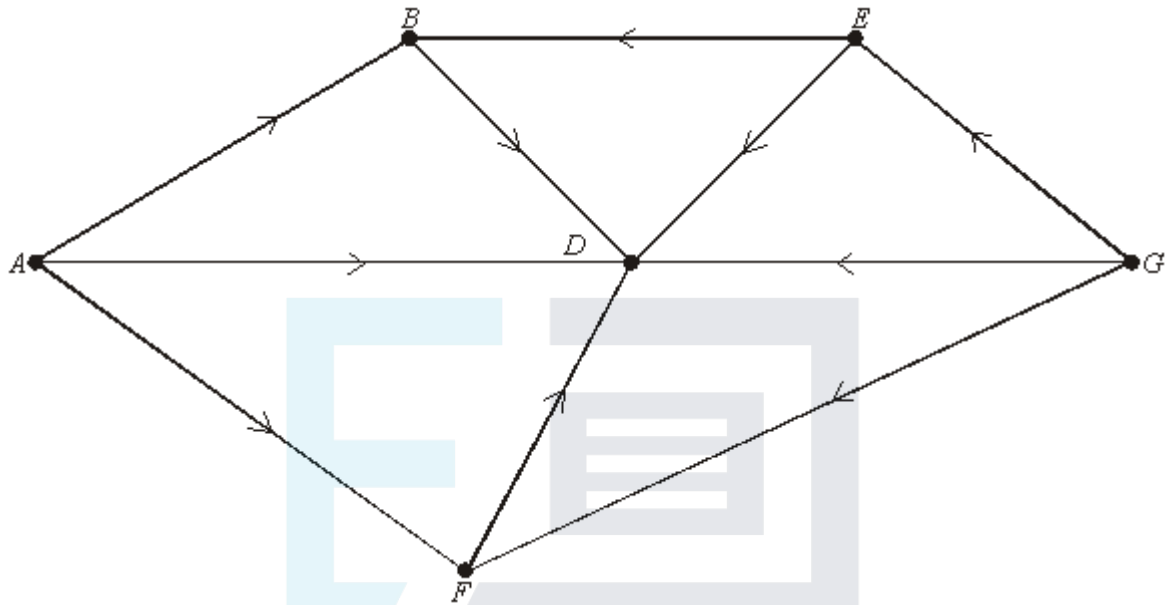
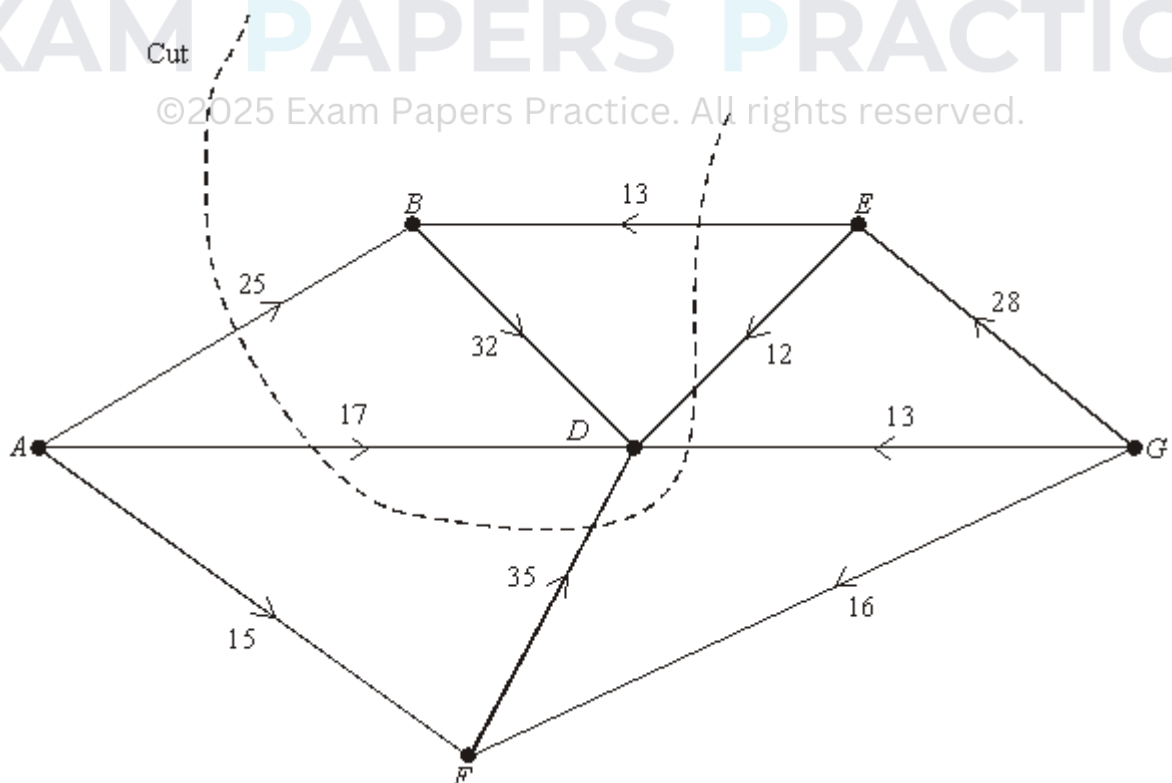


Figure 2



The network shows the routes along corridors from the playgrounds A and G to the assembly hall in a school. The number on each edge represents the maximum number of pupils that can travel along the corridor in one minute.



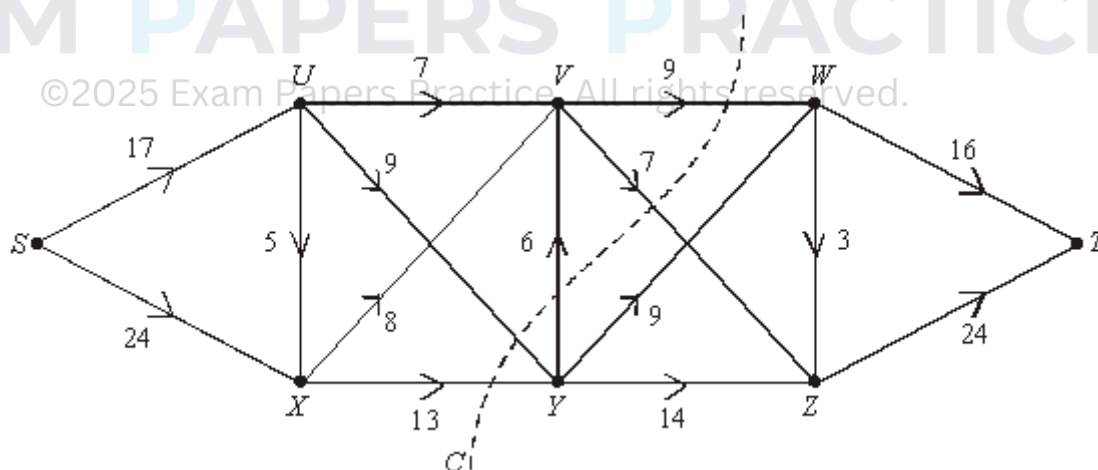
- (a) State the vertex that represents the assembly hall.

(1)

- (b) Find the value of the cut shown on the diagram. (1)
- (c) State the maximum flow along the routes ABD and GED . (2)
- (d) (i) Taking your answers to part (c) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow through the network. (6)
- (ii) State the value of the maximum flow and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow. (2)
- (iii) Verify that your flow is a maximum flow by finding a cut of the same value. (2)
- (e) On a particular day, there is an obstruction allowing no more than 15 pupils per minute to pass through vertex E . State the maximum number of pupils that can move through the network per minute on this particular day. (2)
- (Total 16 marks)

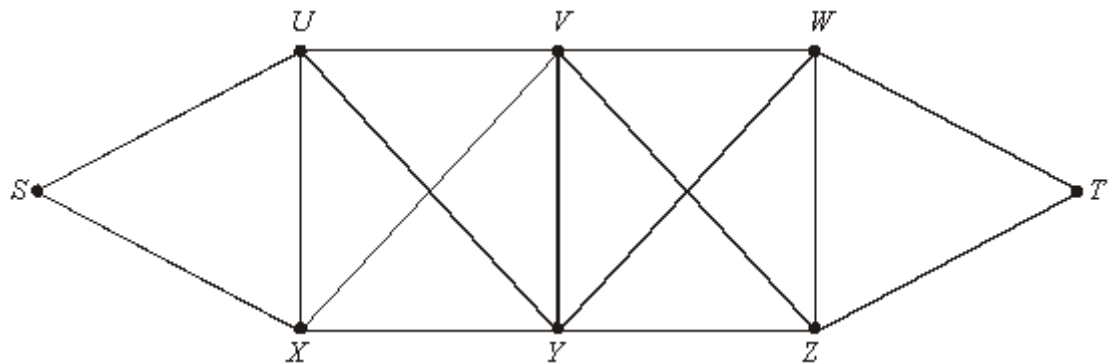
Q18.

The diagram shows a network of corridors in a building. People move steadily along the corridors from the entrance S towards the exit T . The weights on the edges show the maximum number of people per minute that can move along each corridor.



- (a) (i) Find the value of the cut marked C on the diagram. (1)
- (ii) Use your answer to part (a)(i) to make a deduction about the maximum flow of people through the building. (2)
- (b) State the maximum possible flows along the routes $SUYWT$ and $SXYZT$. (2)

- (c) (i) Taking your answers to part (b) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow from S to T



Route	Flow
SUYWT	
SXYZT	

Figure 1

(6)

- (ii) State the value of the maximum flow and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

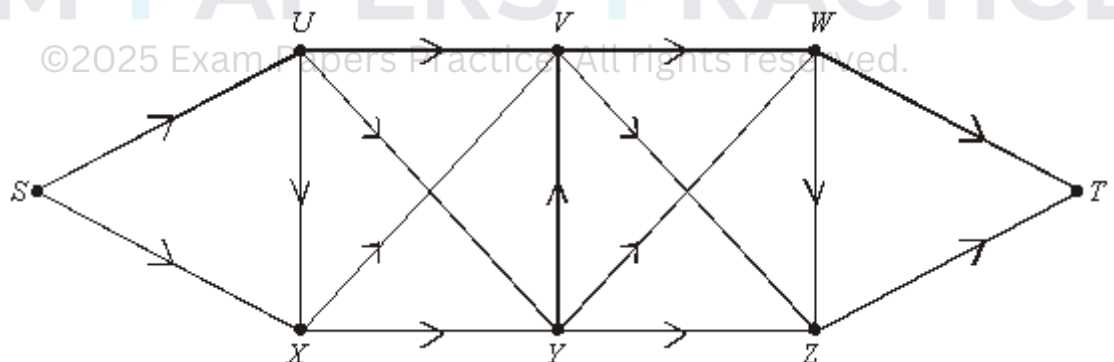


Figure 2

(2)

- (iii) Find a cut through the original network that has value equal to this maximum flow.

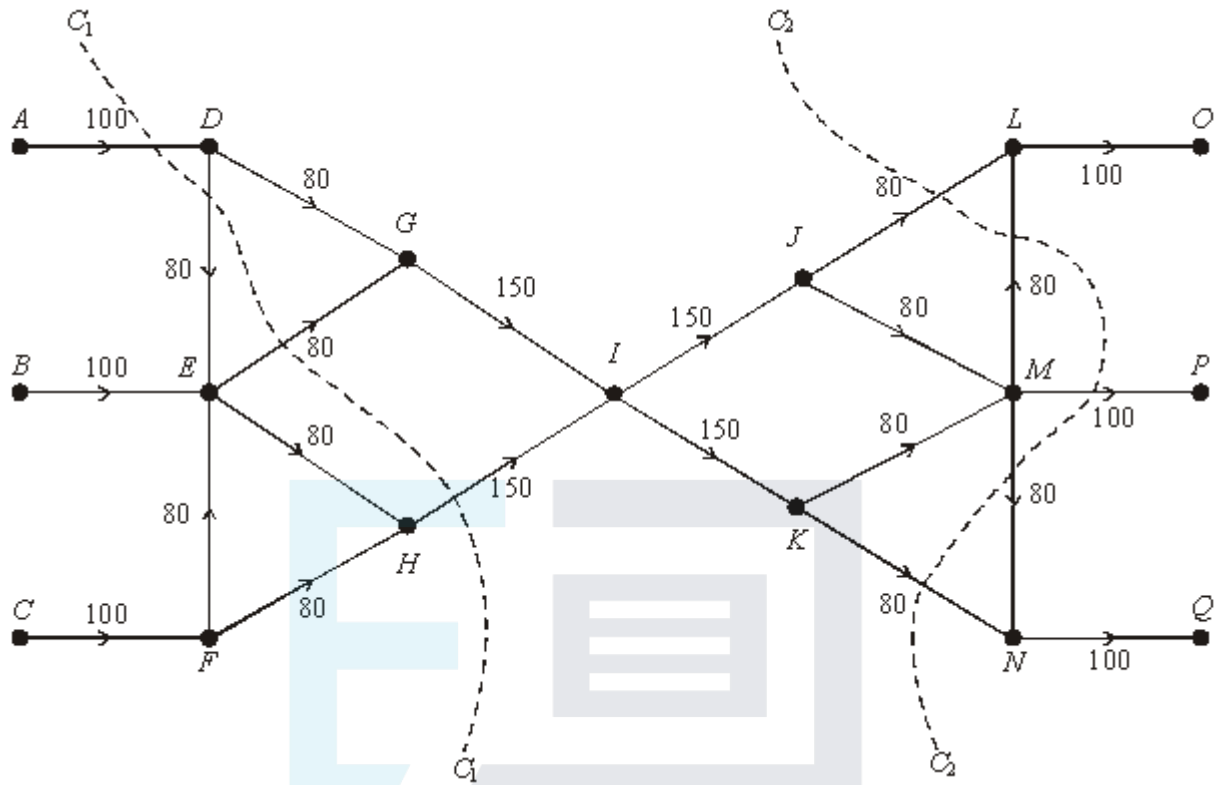
(2)

(Total 15 marks)

Q19.

The following diagram shows the numbers of pupils who can pass along various corridors of a school building in one minute in the event of a fire drill. The classroom areas are A , B

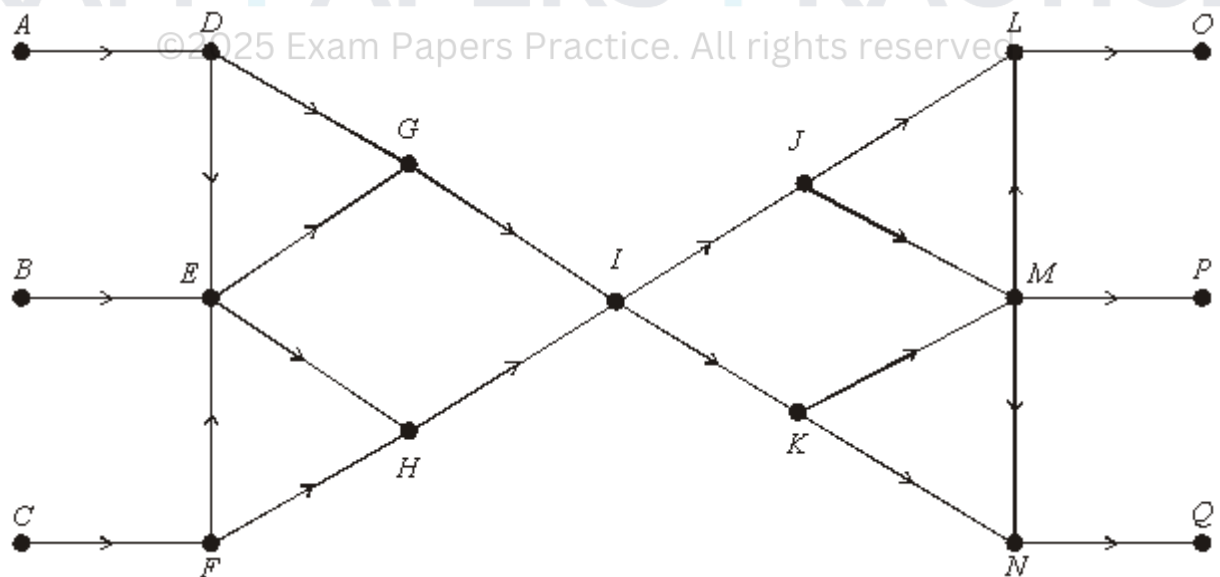
and C . The fire exits are O , P and Q .



(a) Find the values of the cuts C_1 and C_2 .

(2)

(b) On the figure below, show a flow of 300 pupils per minute through the network.



(3)

(c) (i) On a certain day there is a restriction at vertex I of 120 pupils per minute.

Find the maximum flow on this day.

(1)

- (ii) On another day the junction at I is clear but there is a restriction at vertex H of 120 pupils per minute.

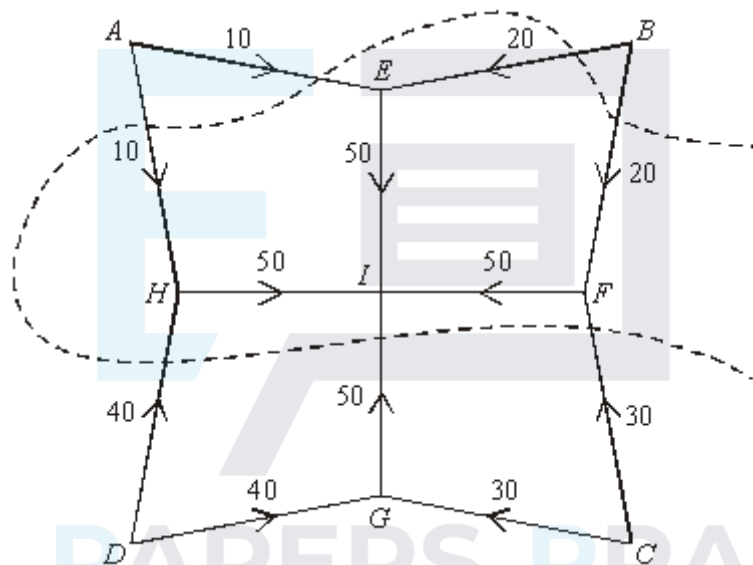
Find the maximum flow on this day.

(2)

(Total 8 marks)

Q20.

The following diagram represents a network of pipes used in a water system in a garden. The number on each arc represents the maximum volume of water, in cubic centimetres, that can flow along the corresponding pipe in one second.



- (a) State the vertices that represent the sources. (1)
- (b) State the vertex that represents the sink. (1)
- (c) Find the value of the cut X . (2)
- (d) (i) On **Figure 1**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

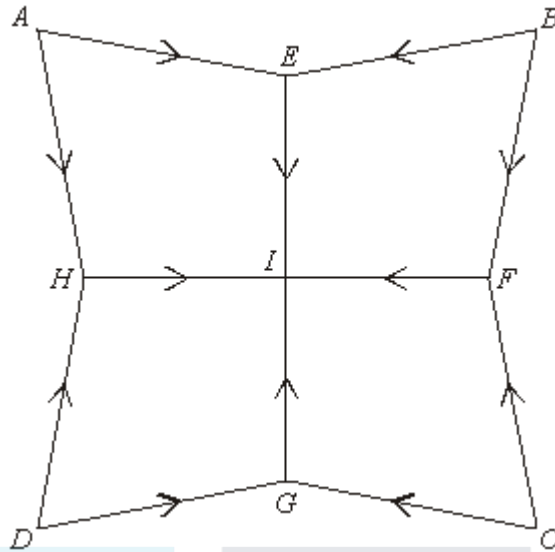


Figure 1

- (4)
- (ii) State why this flow is maximum. (1)
- (e) On a particular day there is a restriction at vertex E which allows a maximum flow through E of 20.

On **Figure 2**, find, by inspection, the maximum flow through the network on this day.

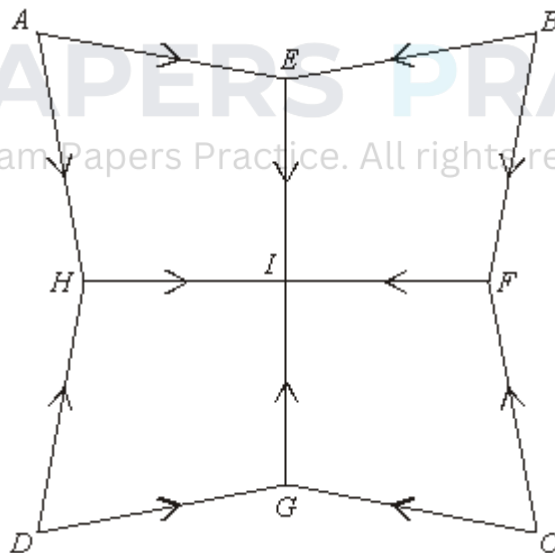


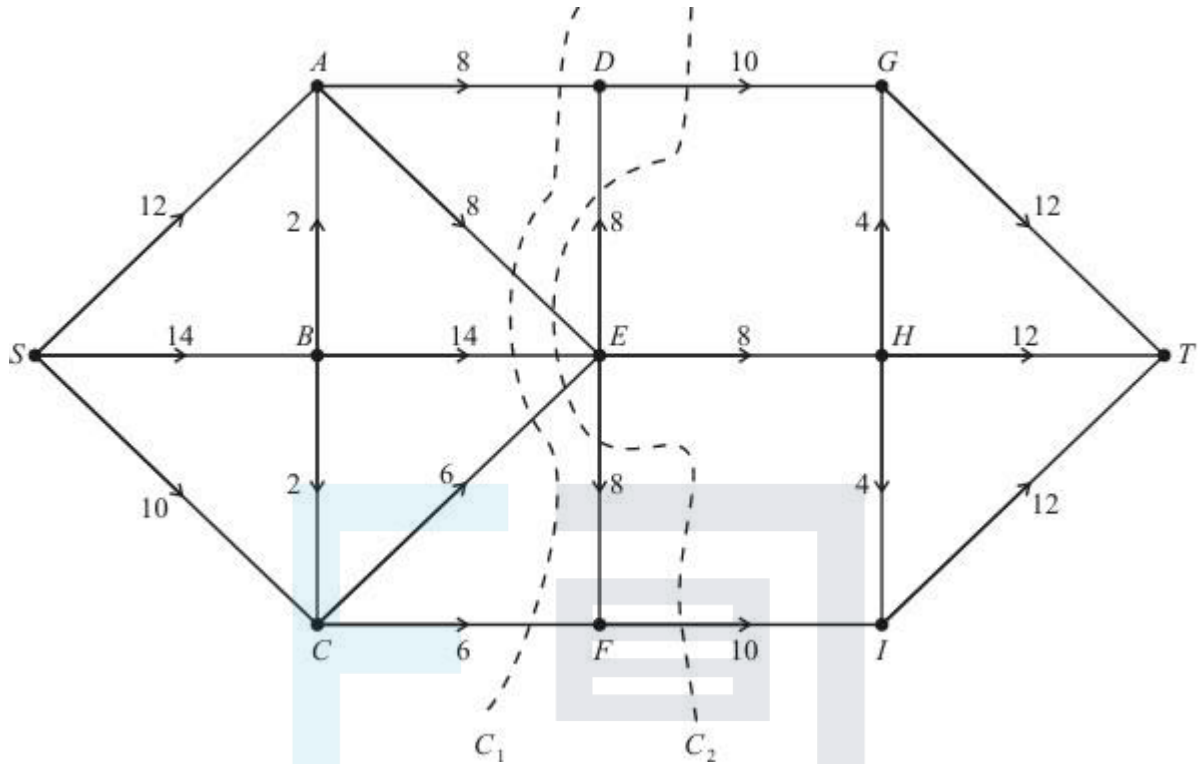
Figure 2

(2)
(Total 11 marks)

Q21.

The following diagram shows a network of drainage pipes on a housing estate. Vertex S is the highest point on the estate and vertex T is the lowest point on the estate. The numbers on the edges show the maximum volume of water, in cubic metres, that can flow

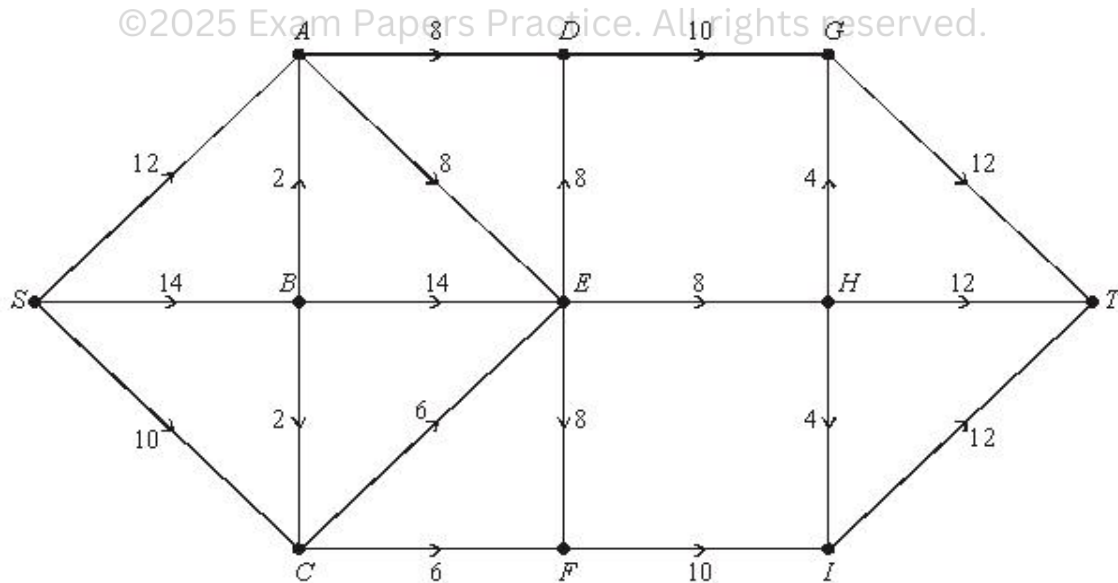
along each section of the pipe in one minute.



- (a) Find the value of the cuts marked C_1 and C_2 on the diagram.

(3)

- (b) On the diagram below, starting from a position of zero flow, use flow augmentation to find the maximum flow of water through the drainage pipes.

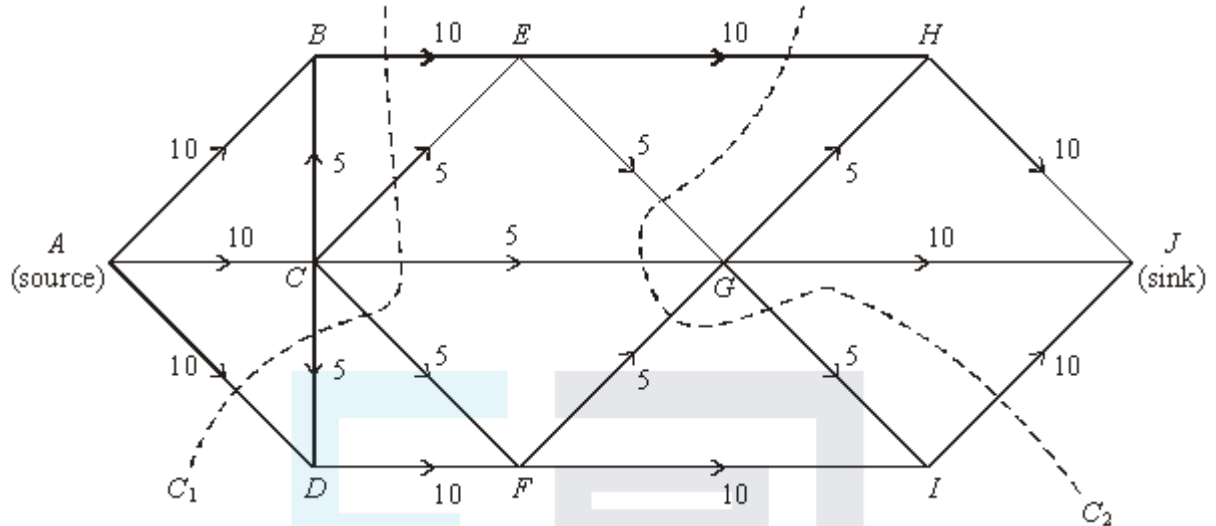


(7)

(Total 10 marks)

Q22.

The following diagram shows ten vertices connected by a number of arcs. The number on the arcs represent the maximum flow along each arc.



- (a) Find the values of the cuts C_1 and C_2 .

(2)

- (b) Find, by inspection, a flow of 30 units and illustrate it on **Figure 1**.

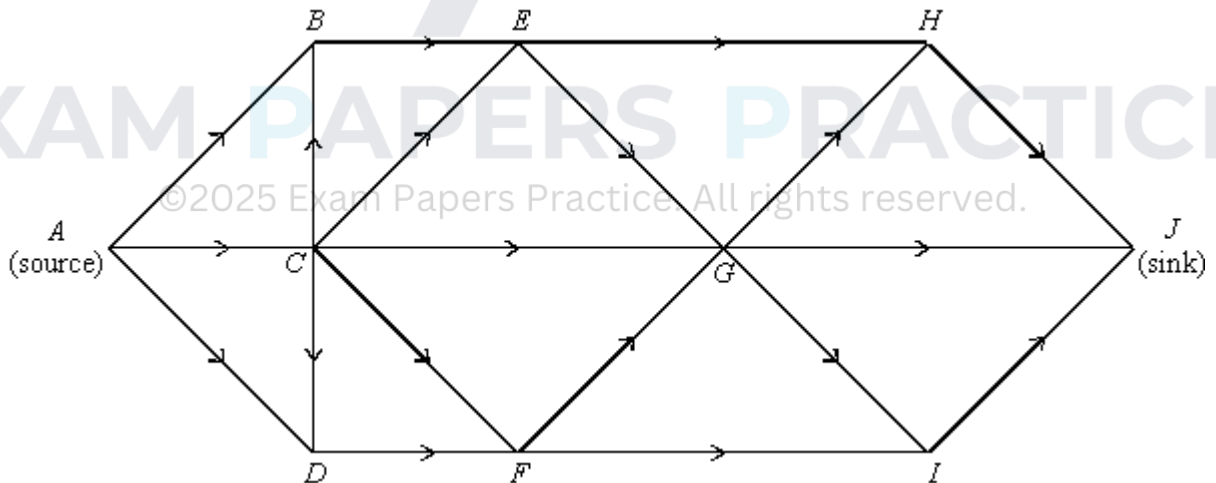


Figure 1

(2)

- (c) On a certain day, the section GJ is blocked so that the network is as now indicated on **Figure 2**.

Find, by inspection, the maximum flow on this day and illustrate it on **Figure 2**.

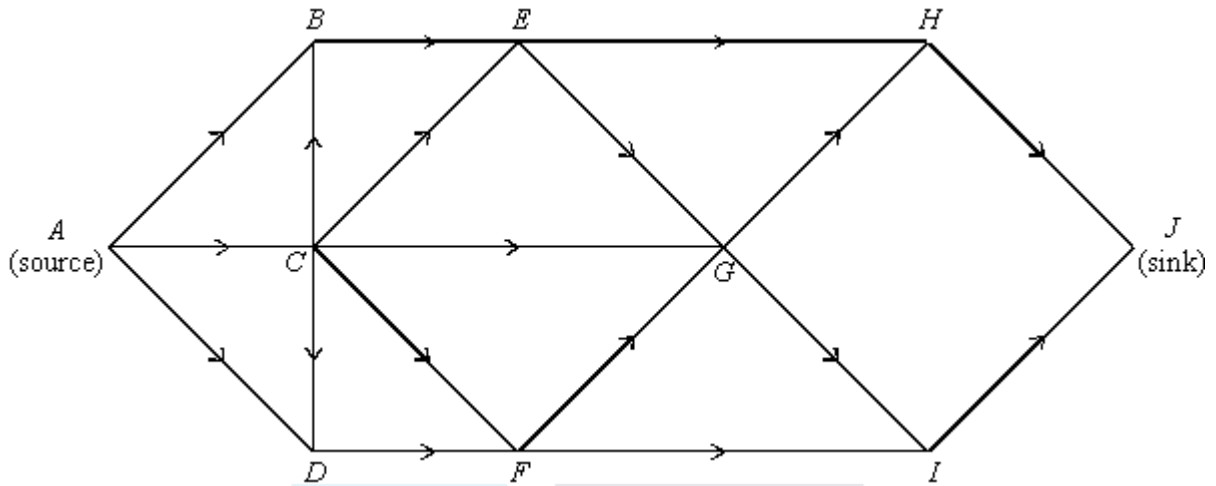


Figure 2

(2)

- (d) On another day, GJ is working normally but EH is now blocked, as shown in **Figure 3**.

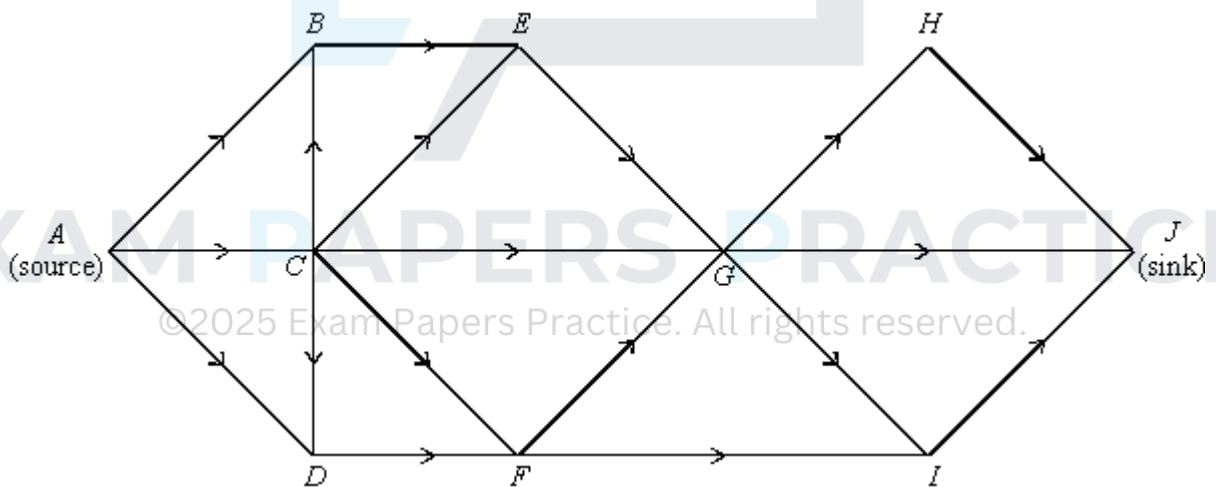


Figure 3

- (i) Find, by inspection, the maximum flow on this day and illustrate it on **Figure 3**.

(2)

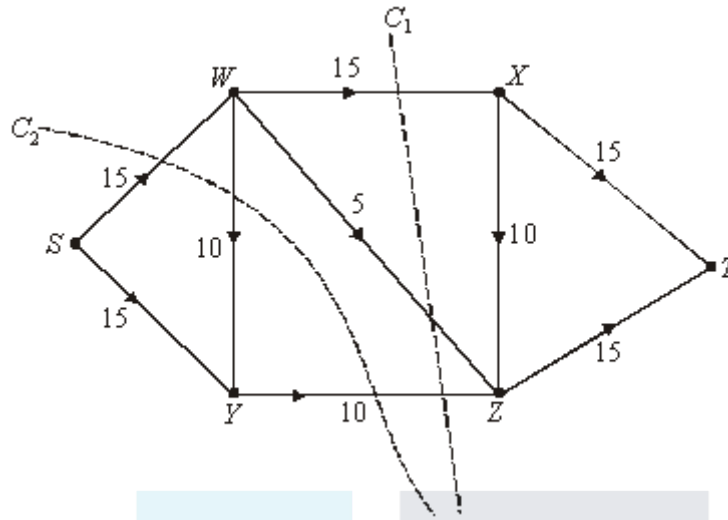
- (ii) Show that the flow obtained in part (d)(i) is maximal.

(2)

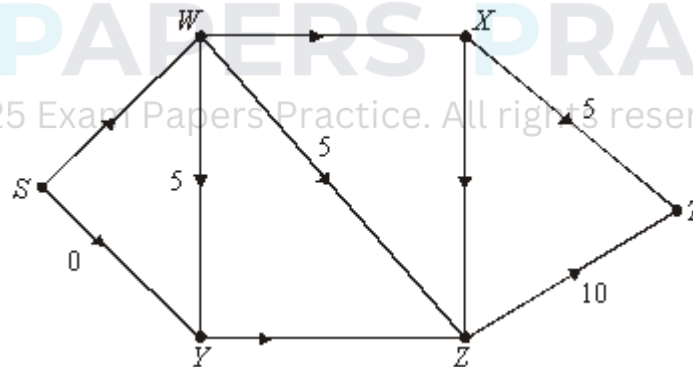
(Total 10 marks)

Q23.

In the network shown below, S is the source, T is the sink, and the number on each arc indicates its capacity:



- (a) Find the value of:
- (i) the cut C_1 ; (1)
 - (ii) the cut C_2 . (1)
- (b) The diagram below shows a flow from S to T in which the individual flows in four of the arcs have been omitted.



- Calculate the flows in YZ , XZ , WX and SW . (3)
- (c) Find two flow-augmenting paths, each of value 5, which together can be added to the flow illustrated in part (b). (4)
- (d) Explain how you know that no additional flow-augmenting paths exist. (2)
- (Total 11 marks)**